

Aspects of Classical Chinese grammar

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September 3, 2026

Chapter 1

Introduction

1.1 The language

This note is about Classical Chinese, the high variety of more than two millennia of diglossia in China. The language is known natively (in Mandarin Chinese) as 文言 ('lit. cultured speech') or sometimes 古文 ('lit. ancient articles') or 古汉语 ('lit. ancient Chinese'). Note that there were several stages of the development of Chinese and Classical Chinese is mostly (but not completely) based on Old Chinese (TODO).

The language is sometimes known as *Wen-li* by Western missionaries, especially in Bible translation. This seems to be a misunderstanding of the word 文理, which is a nominal compound and means rhetorics (i.e. 文) and meanings (i.e. 理) of literature works. An educated person therefore would be described as “通文理” ('fluent in rhetorics and meanings'). Such a person of course would have decent understanding of Classical Chinese, and hence 文理 was probably mistranslated as “Classical Chinese”, although the word was not natively used to refer to the latter.

The word 文言 is sometimes translated to *Literary Chinese* in English. This translation can be misleading. Some written materials in Chinese were definitely *not* written in 文言 i.e. Classical Chinese, like novels like *Journey to the West* or *Dream of the Red Chamber*: they were written in earlier forms of Mandarin. Buddhist scriptures translated to Chinese are more or less Classical but are also definitely influenced by the grammar of Middle Chinese. The non-Classical written materials, despite being important both culturally and linguistically, are beyond the scope of this work.

Since there was no attempt at explicit and systematic grammatical standardization (§ 1.4.1), the prescriptive authority of Classical Chinese was a collection of canonical literature works consensually regarded as classical (§ 1.2). The whole canon was finished before the collapse of Han and therefore falls under the term Old Chinese. Both temporal and regional variances can be observed in Old Chinese texts, though, and not all varieties contribute to Classical Chinese equally. (TODO: specify which varieties were most influential.)

1.2 The Classical period and texts

Based on philological evidence, the early history of Chinese is commonly divided into an Old Chinese period containing both the Pre-Classical period and a Classical period from the Spring and Autumn period (§ 1.2.1) to late Han dynasty (§ 1.2.2), and a Middle Chinese period.

We consider texts before the Spring and Autumn period Pre-Classical, despite their classic status in a Confucian context. This periodization is consistent with traditional philological scholarship in China: Han Yu, renowned for reviving Classical prose, famously described the *Book of Documents* as 诘屈聱牙 ('twisted, hard to pronounce') in his 进学解 (*Analysis of academic advancement*). The *Book of Change* similarly is not comprehensible to a scholar trained only in Classical Chinese and requires commentaries to be understood. The *Book of Odes* includes poems spanning a long period of time from early Zhou to Spring and Autumn, and does not have a single, coherent grammar. Thus, these texts are excluded from the scope of this work.

This work attempts to analyze the shared grammatical traits of these admittedly highly diverse texts. This resembles the stated goal of Huddleston and Pullum (2002), which is to analyze *International Standard English*, i.e. the shared core of all important contemporary varieties of English. The existence of such a shared Classical grammatical core should not be taken for granted and can only be proven by demonstrating that most texts considered Classical are analyzable under the framework of this work.

The mental grammar of an author of post-Han 文言 prose almost certainly differed from any possible grammar of Classical writers. A radical example is texts from the Republican period of modern Chinese history, when official documents were written largely in a pseudo-Classical style: the abstract syntax behind these documents is almost certainly within the framework of modern Mandarin, with the possessive marker 的 replaced by 之 and the relative marker 的 in headless relative clauses replaced by 者 (不服管教的 'who does not conform to instructions' is replaced by 不服管教者). What is Classical in these documents is primarily the superficial forms. Still, because of the highly formulaic nature of these documents (for instance, usually no sentence final particle appears, and valency alternation besides the passive is discouraged), readers of these texts still benefit from studying the grammar of Classical Chinese. More details can be found in § 17.2.2.

1.2.1 Spring and Autumn and Warring States

The majority of texts that shaped Classical Chinese prose were written in a time when Zhou was already substantially weakened. This period that witnessed prolificacy of Old Chinese works can be divided into two periods: the Spring and Autumn period which was filled with chaotic (but not intense) wars between numerous dukedoms, and the Warring States period which observed intense wars between seven major states, resulting in a unified Qin empire, which soon broke down because of resistances to its barbaric policies and eventually was superseded by Han dynasty (§ 1.2.2). The language of this period diverges tremendously from the pre-Classical period. For example, the copula 惟 had died out of use and the copula construction had been largely replaced by the nominal predication construction (§ 2.2.1). The conjunction 而 is never seen in pre-Spring and Autumn texts but had already made its way into the *Analects*. The lexicon also underwent huge changes.

Box 1.1: Lexicon change

List some lexicon changes.

There are clues suggesting regional variances. Students of Confucius noticed that when he recited Classical texts and presided rituals, he used 雅言 or 'elegant speech' (*Analects* 7:18). This suggests a possible diglossia at as early as Confucius's own age,

with the “elegant speech” conceivably being the language of intellectuals of Zhou Dynasty. Comparison between the language of Classical prose and the language(s) of poetry shows the relative homogeneity of the former, while the latter both demonstrate divergence from the language of prose and regional differences.

Box 1.2: Poetry and prose

This is presumably due to how the texts were transmitted. It is likely that they were passed by recitation, and regularization happened to prose when there was a predominant dialect, while the prosody and rhyme structures of poems efficiently locked them to their original forms.

The language of 楚辞 (*Verses of Chu*), for example, has the following differences from the language of prose. The first is a Kra–Dai substrate.

Box 1.3: Chu dialect

Find references.

The language of the *Odes* also seems to differ slightly from prose Classical Chinese. Dialectal differences have also been observed within the *Odes* (List et al. 2017).

1.2.2 Han dynasty

The last batch of uncontroversially classical works were composed during Han dynasty, among them the most important being *Records of the Grand Historian*. The language of *Records of the Grand Historian* shows notable but largely qualitative differences compared with earlier historical works, the most important one being 《左传》. Notable changes include more pre-verbal adverbials, reduction of prepositional verbs, regularization of constituent orders, and also proliferation of disyllabic words. It is therefore suggested that Han dynasty texts and pre-Qin texts reflect two stages of post-Zhou developments of Chinese, although the change was definitely not as radical as the change from the *Documents* to Spring and Autumn texts (He 2005, pp. 260-264).

The great historical work 《史记》 (‘lit. historical records’), often known as *Records of the Grand Historian* in English (a translation of 太史公记, the earliest known title of the work), laid the paradigm of official historiography of all Chinese dynasties after Han. The author 司马迁 *Sima Qian* is known as the *Lord Grand Historian* or 太史公. 太史 ‘grand historian’ was in charge of administering archives and making calendars.

Box 1.4: List of Classical texts

- Pre-Qin, and Han
- 唐宋八大家?

1.2.3 Post-Classical periods

The end of Old Chinese – and hence the end of the classical period – is marked by the collapse of the case inflection in the personal pronoun system, the emergence of 是 as a copula (and not just a demonstrative), the appearance of the disposal construction (i.e. the 把 construction) and the so-called long passive construction.

Box 1.5: References for Middle Chinese and modern Mandarin

- James Huang
- etc.

Expectedly, despite purification attempts, vernacular elements made their ways into not only administrative documents but also pure literature and scholar works. Classical Chinese or 文言, in the broadest sense, is a term that covers all genres whose grammars are roughly based on the Old Chinese canon but may have a variety of innovations.

Box 1.6: Late regularization attempts

韩愈、唐宋八大家、因明学

1.3 Theoretical framework

The theoretical framework of this work is strongly inspired by Distributed Morphology (for an introduction see Siddiqi (2009)) plus Cartographic Syntax (e.g. Cinque (1999)), although I intentionally choose to reuse the terminology in descriptive grammars. We will see that the structuralist tradition of Chinese linguistics agrees incredibly well with the generative framework.

1.3.1 Lexicon, grammar, wordhood

The architecture of a language is assumed to be in line with the basic assumptions of Distributed Morphology. The so-called “lexicon” consists of a list of roots (List A), each of which is only realizable in combination with grammatical formatives (corresponding to certain syntactic positions) (List B), and grammatical objects of arbitrary sizes with custom meanings (List C).

List B dictates lexicalization of grammatical properties of forms (i.e. subcategorization; cf. § 8.2.2.2), which is related to wordhood (§ 1.3.5). We note that this does not entail the necessity of a lexicalist grammatical framework where structure building mechanisms for words and phrases are categorically different (cf. Bruening 2018).

List C dictates idiomization, which may also be known as semantic lexicalization. The distinction between (syntactic) lexicalization and (semantic) idiomization is discussed in e.g. McInnerney (2023).

Lexicalization and/or idiomization of a complex structure may eventually lead to the collapse of its internal makeup, causing (*syntactic*) fossilization. An example is the collapse of verbs taking complement clauses into compound verbs. Fossilization may end up producing a single *root*, of which only diachronic analysis is possible. Syntactic fossilization is important in Classical Chinese due to its long history.

Parts of speech labels hold different meanings in different contexts. The noun may just be defined as the head word of a noun phrase; hence notions of “verbs used as nouns”. Alternatively, people refer to *form classes* in the lexicon that share similar lexicalized properties as parts of speech.¹

¹This obviously begs the question: where does the rigidity of these form classes come from? Why can’t, say, a Latin noun have no gender feature? This may be explained by cross-linguistic variances in List A: we may assume that the nominal categorizer is gendered (Kramer 2015). Alternatively this

This second definition of parts of speech sees more cross-linguistic variances. To say that the Latin word *canis* is a noun means that *canis* belongs to a form class in which every member usually appear as the head of a noun phrase (NP), carries an inherent gender feature and a number feature, and that its inflection pattern follows one of Latin nominal declensions. (It's easy to imagine how such a form looks like in Distributed Morphology.) To say that *dog* is an English noun means something else: it means the root *dog* regularly appears in the nominal modification construction (as in *a [dog]_{nominal} (not NP) tag*), which is not attested in Latin.

1.3.2 High-level constituents; grammatical relations

The analyses in § 2.2 and § 2.3 are inspired by the extended CP and DP structures proposed in Cartography. To avoid confusion caused by technical terms in generative syntax, I intentionally use terms like *sentence*, *nucleus clause* and *argument structure* in place of CP, TP and vP. Similarly we talk about *noun phrases* instead of DPs.

Concepts like SpecTP and SpecvP, in a descriptive context, correspond to concepts like *clausal subject* and *pivot in the argument structure* (§ 2.2.4). Structural positions definable at the vP level we call argument slots below (the much more frequently used term in generative syntax is thematic roles; the correspondence between traditionally semantically motivated thematic role classification and the light verb shell has been extensively discussed, as in the literature on UTAH), and structural positions definable at the TP level we call *grammatical functions*. The “core” of CPs and DPs (i.e. the roots at their centers plus the categorizers) – instead of the grammatical markers – should be known as heads.

In CPs and DPs recurring patterns or *constructions* and *grammatical relations* in them can be observed: subject-predicate and verb-object relations, subordination and coordination structures.² The fine details of the light verb shell I replace by concepts like *DO clause* or *CAUSE clause* (§ 2.2.2.2).

Grammatical markers can be distinguished from content roots and words in that grammatical markers are from a closed class and mark grammatical categories and functions in certain constructions, and have no meaning outside these constructions. This often means that forms containing grammatical markers that seemingly semantically make sense are not acceptable (e.g. § 2.2.3).

The correspondence between descriptive notations and theoretical linguistic concepts has been demonstrated in Deng (2010), which shows that Minimalist generative syntax and the constituency-based analysis seen in the Chinese linguistics tradition outlined in § 1.4.3, which appears strongly inspired by American structuralism exemplified in Huddleston and Pullum (2002), are compatible to each other, with the former being a more concise form of the latter and the latter being a logical consequence of the former under certain assumptions.³ Further,

may be explained by a language-specific well-formedness condition. The competition between the two analyses is a part of the wider macroparameter vs. microparameter debate, which we do not touch in this work.

²Here the term is used *without* the implication that a construction is somehow understood as a whole and its internal structures should not be further analyzed, contrary to the fundamental hypothesis of e.g. various Construction Grammars.

³Note that Deng (2010), Huddleston and Pullum (2002) and works in § 1.4.3 are all lexicalist, which we have argued is not necessary to account for phenomena purportedly supporting the lexicalist hypothesis. These works are also based on the more traditional X-bar theory: the syntactic differences between the subject and the object, for example, are explained by the difference between specifier and complement. In Cartography, what is recognized as a complement may be a lower specifier: thus [VO]_v is replaced by a “transitivity phrase” [O [TRANS √ V]_{TRANS}]_{TRANS} (Siddiqi 2009), where the category-less verb root is first combined with a transitivity functional head and then combined with the object. The resulting tree diagram is quite like the X-bar scheme before, say, the DP hypothesis: the only difference being that there are many X'-like projections, all of which may appear in other constructions. For in-

as shown in § 1.3.3, the constituency-based analyses are consistent with the Basic Linguistic Theory (the de facto unified framework in modern linguistic description of underdocumented languages).

The three approaches can be seen as three “representations” of the same grammatical complexity class, and their differences are mostly notational.⁴ Which framework to use is determined by the properties of the language. For instance, although the definition of the verb phrase does not alter the grammar system substantially, Dixon’s definition works more smoothly for a language with a lot of auxiliaries but rather infrequent subject-sharing coordinations. It turns out that for most constructions, the method in Huddleston and Pullum (2002) is a good choice for Classical Chinese.

We note that we do not consider adjunction an independent structure-building process that is different from Merge. A detailed discussion can be found in sections on the argument/adjunct distinction Ch. 2.

1.3.3 Dependency

Grammars can be written in terms of *dependency relations*, instead of *constituency relations*. The two however are largely equivalent (Boston, Hale, and Kuhlmann 2009). In his Basic Linguistic Theory, Dixon (2009) fervently argues against constituency analysis (and also other aspects of generative syntax) and advocates for a “flat” constituency structure, possibly with the levels of clause and noun phrase only, where the rest of the grammatical information is represented by dependency relations (e.g. Dixon 2009, p. 49). Yet the primacy of noun phrases and clauses in syntax is indeed acknowledged by generative syntax (for instance, they are *phases*; it is also not uncommon to see terms like *CP syntax* or *DP syntax* in generative works).

The binary constituency relations in generative syntax and in American structuralism are not arbitrary and have consequences. For instance, when applied to the argument structure, they are related to properties of arguments in valency alternation (§ 2.2.4.2), and the subject-predicate binary division is directly related to the clausal pivotal status of the subject. These phenomena of course have to be taken into account by Basic Linguistic Theory, and labels like “clausal pivot” and “surface S, deep O” have to be attached to dependency arcs between the verb and the arguments, essentially labeling the “distance” between the two. As a parallel, the ancient Indian grammarian Pāṇini initially proposes a grammatical framework in which arguments are all equal, but later commentators still effectively set up a pivot position in the argument structure.⁵

stance, the English *nominal* mentioned in Huddleston and Pullum (2002), which is often recognized as N’, can actually be a modifier in the DP, as in *a [Fuji apple] tree*. Deng (2010) shows that the subject-predicate structure or the verb-object structure can all be derived from the traditional X-bar scheme: the Cartographic approach pushes this decomposition even further and demonstrates that the traditional X-bar scheme itself can be further decomposed into a series of functional heads and roots.

⁴A more important controversy is the mental status of grammar. Generativism treats grammar as a semi-autonomous component of human cognitive abilities. But it has been argued that grammatical constructions originate from domain-general cognitive abilities, and there is actually no such thing as an autonomous mental grammar. Detailed discussions on this topic are far beyond the scope of this work, and can only be finally settled down with the assistance of neurological studies. Here we just note that currently no comprehensive description of a language has been successfully attempted under this line of thinking. On the contrary, in physicists’ terms, the grammatical framework adopted here is at least a good *effective theory*. So-called functional-typological grammars do not differ significantly in the way they carry out analyses from structuralist works.

Another topic is whether natural language-like “grammars” appear in systems that are demonstrably independent to the human neurological linguistic capacity, which may also challenge the domain-specific status of the latter. This is partly discussed in § 3.1. Note however that such discussions are largely irrelevant to linguistic description, for reasons mentioned above.

⁵The main difference between Pāṇini’s treatment of Sanskrit and modern linguistic theories is that

Other phenomena that can be attributed to binary constituent structures include ordering of tense and aspect auxiliaries and adverbs à la Cinque, attested in languages completely unrelated to the Germanic or Romance languages that have been thoroughly investigated in Cartography (e.g. Grimm 2021, pp. 166-167), and also available clause sizes: it is rare that in a language with tense, aspect and modality marking, there exists a clause type with discourse devices but no tense, aspect and modality marking, which can be explained simply by the fact that CP is built on TP.

1.3.4 On the term *phrase*

Besides the theoretical problems outlined above, there are also some minor, largely notational inconsistencies between the grammatical theories mentioned here. One such inconsistency is the definition of the *phrase*. Dixon (2009) calls the main verb plus auxiliaries – without any argument – the *verb phrase*. This notion of verb phrase has a generative syntax counterpart: the CP phase. In a CP, arguments are phrases themselves, and the tense, aspect and modality categories marked by the auxiliaries are thus in some senses closer to the main verb. On the other hand, the term *verb phrase* generally means the verb plus internal arguments in constituency-based analyses (as in e.g. Danckaert 2017).

1.3.5 Wordhood

Another problem is the definition of *word*. Since we reject the lexicalist hypothesis, we need to distinguish phonological wordhood from morphological wordhood (where the boundary of the morphological template is the boundary of the morphological word).

We first note that it is not possible to define wordhood purely on (abstract) syntactic structures. We can define a word to be a very small constituent: if it is impossible to infer the argument structure of a compound verb in a given language, then we conclude that the two branches of the compound are not categorized, and therefore the compound is a rather small constituent and hence a word. But in this way *sinned* is *not* a syntactic word as it involves a clausal category (i.e. the past tense). For *sinned* to be a syntactic word, syntactic wordhood can be based on Dixon’s verb phrase, and inevitably *have been eating* is a syntactic word.

Therefore, (abstract) properties of structure building processes cannot be the only criterion used in defining wordhood.⁶

Morphological wordhood, i.e. grammatical wordhood, corresponds to post-syntactic aggregation of roots and formatives in Distributed Morphology, guided by List B. Its traces can be found in existence of morphological templates that can’t be motivated by purely combinatorial syntax, boundaries of forms in phonology that lack pure prosodic motivation, etc. Further, the way List B works means wordhood is strongly related to lexicalization, as the inability

Pāṇini apparently treats all dependency relations equally and there is, for example, no concept of the pivot or the “external argument” of a clause. This is however modified in the commentaries on his *Aṣṭādhyāyī*, which explicitly allow an argument to be promoted to the agent position because of the intentions of the speaker (Keidan 2017). The agent position thus becomes a subject position at least in the argument structure, consistent with modern practices (§ 2.2.4.2).

The Paninian tradition therefore is extremely close to modern linguistic description practice; the most important difference probably is that modern linguistic description, practically, may even be less rigorous than *Aṣṭādhyāyī*, because of possible competing “mind grammars” among speakers with mutual intelligibility or even within the mind of one speaker, and also the fact that a description as detailed as *Aṣṭādhyāyī* requires corpus data whose quality and quantity exceed the capacity of most field linguists.

⁶This is to be contrasted with the criteria distinguishing compounds and nominal complementation in Huddleston and Pullum (2002), which does rely heavily on abstract properties of perceived word-like and phrase-like structure building processes.

for a transitive verb to have no object is attributed to its stem cannot be spelt out as a grammatical word without a transitivity head (see Siddiqi 2009 for details). Therefore, the lexicon (understood as List B) is often said to contain *words* or *lexemes* (the latter term assumes a clear inflection/derivation distinction; see below). But note that a form with lexicalized grammatical properties may only be a half-word, like English neo-classical combining form *geo-* (which prefers an adjective-like position but is not a word).

The distinction between derivation and phrasal grammar follows from the definition of word. Ideally, the direction of inference should not be reversed: a structure building process being structurally limited (e.g. impossibility of coordination) should not automatically qualify it as a derivation process (though in the existing literature, the derivation/phase grammar distinction is often dealt in this way). These structural criteria are also rarely decisive: the form *pre- and post-revolutionary* obviously mixes phrasal coordination with affixation, but the latter is prototypically derivational.⁷

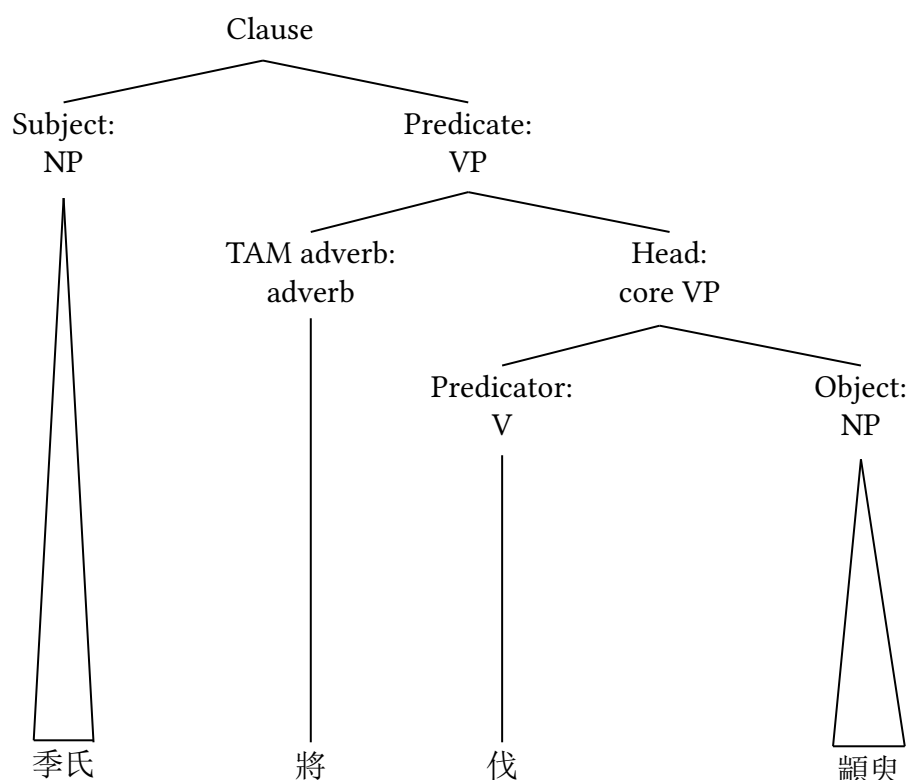
The derivation/inflection distinction involves multiple parameters that do not always coincide. The distinction may be defined based on abstract structural criteria (thus tense marking is inflectional while argument structure alternation is perhaps not); this inevitably has to rely on a pre-defined concept of wordhood (or otherwise auxiliary verb constructions have to be considered inflectional; cf. the same problem faced in syntactic definition of wordhood). The distinction sometimes just means the synchronic/diachronic distinction: derivation processes are supposed to be less productive, a criterion largely consistent with the structural criterion (as a smaller grammatical unit also tends to fossilize faster), but productive derivation processes exist. The most uncontroversial definition of the derivation/inflection distinction relies on the morphological template of words, when the inflectional ending can be relatively clearly set apart from the stem (this, though, means stem alternation marking of aspect in Latin is derivational).

1.3.6 Notations

We have mentioned that the framework of Huddleston and Pullum (2002) seems to be a good starting point of the description of Classical Chinese. An example of the tree diagram representation of the constituency analysis (e.g. Huddleston and Pullum 2002, p. 954 [9]) is shown in Fig. 1.1. This representation however does not clearly distinguish function words from content words.

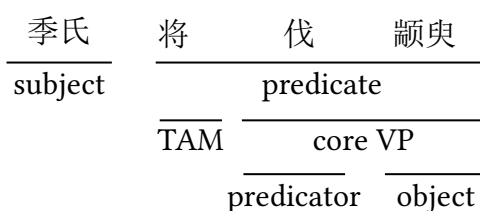
⁷In this concrete example, we consider *pre-revolutionary* a grammatical word because phonologically the boundary between *pre-* and *revolutionary* doesn't matter: the form is read as a single phonological word and there is a single stress. This is hard to explain if there are two grammatical words in it.

Figure 1.1: Tree diagrammatic analysis of 季氏將伐顓臾 ‘Jishi (a noble family which controls the politics of the Kingdom of Lu) is going to invade Zhuanyu (a vassal state of Lu).’ (*Analects*)



Researchers in modern China conventionally draw lines below constituents with labels to represent constituency relations. Fig. 1.2 contains exactly the same information in Fig. 1.1 and is more compact. It also highlights the lexical/functional distinction. For instance, if 將 were a future tense particle and not an adverb, we could refrain from giving a label to it (and eliminating the “TAM” layer in Fig. 1.1), implying that it is a grammatical marker and not a (content) constituent of the predicate.

Figure 1.2: More compact version of Fig. 1.1



The problem of Fig. 1.2 is also its compactness: the label of a constituent cannot be too long, or otherwise the whole diagram cannot be fit within one line. In this work, we represent examples in a way that is more aligned to modern description grammars: we place constituents in brackets, and then label these brackets by subscripts. (1) is an example. When we have grammatical markers, we do not put them (e.g. the sentence final particle 矣 in 2) in brackets to highlight the lexical/functional distinction, consistent with the suggestion of Dixon (2009, p. 49).

- (1) [[季氏]_{subject: NP} [[將]_{TAM adverb} [伐
 Ji-family.branch.lineage.name will subdue
 顓臾]_{core VP}]_{predicate: VP}]_{verbal clause}
 Zhuanyu
 ‘Jishi is going to invade Zhuanyu.’ (Analects)
- (2) [[[民]_{subject: NP} [鮮 —object]_{predicate: VP}]_{subject: complement clause}
 people lack
 [久]_{predicate: AdjP}]_{verbal nucleus clause} 矣_{SFP}]_{sentence}
 long SFP
 ‘It has been quite long that people lack this!’ (Analects)

One further issue is phonological transcription of the texts. The standard practice is to apply the Modern Mandarin phonology (romanized in *pinyin*), following the reading tradition (Pulleyblank 1995); this is comparable to how ancient Greek texts are read in modern Greece. Historically, it was also possible to read Classical Chinese with different phonological systems.

This standard practice does not give us much insights into the phonological reality of these texts. Some choose to transcribe texts in reconstructed Old Chinese (Jacques 2022). Details in reconstruction of the phonology of Old Chinese remains a highly debated topic, and no mutually accepted lossless romanization that reflects the historical phonology exists currently.

1.3.7 Procedure of analysis

We adopt a top-down order in writing this book. We start with high-level constituents, i.e. clauses and noun phrases. Grammatical systems are identified when these forms are analyzed, and are introduced in the rest of the sections of § 2.2 and § 2.3. Parts of speech divisions are made based on grammatical evidence and not semantics.

This organization is in contrast with some descriptive grammars, which start with e.g. the nominal and verbal morphology, and then discuss grammatical relations like possession, modification, and argument structure, and then outline the structure of the simple clause and complex clauses. This is not the approach taken here, partly because Classical Chinese is a rather analytic language and there is simply no inflectional morphology to describe, and partly because a well organized study of its grammar has to have the whole syntactic environment in mind: otherwise it will likely collapse into a dictionary of the various usages of function words.

This work also pays attention to the combinatorial possibilities of dependents in clauses and noun phrases and their relative orders and scope, which is often skipped in *ethno-philological* works mainly aiming to prepare the reader to comprehend *natural texts* in the language in question. We aim to study the grammar of Classical Chinese as a machine that takes lexical items and produces *arbitrarily* complex utterances, and to understand the structure of this machine. The approach is admittedly not unproblematic for a dead language, which has no native speaker with acceptability intuitions and may also have strong internal variances historically.

1.4 Previous studies

1.4.1 Native grammatical traditions

Classical Chinese authors had conversations about grammaticality and uses of grammatical particles reminiscent of how English native speakers with some exposure to the study of English grammar but no formal training: “delete the *the* here and your sentence looks more concise”. No attempts were made to establish intermediate concepts between words and utterances, like structural templates of phrases or grammatical relations, and to organize the grammar as a machine producing acceptable utterances: discussions on grammatical topics were either for education or for rhetorics.

The grammatical awareness of Classical Chinese authors was somehow comparable to what an ancient Roman grammarian or *grammaticus* did, who set his main role as a secondary educator, refrained from analyzing some sort of “underlying” or “internalized” system behind the surface forms and was satisfied by mostly surface-oriented patterns, and would not set up any intermediate concepts between the word and the utterance (Matthews 2019, pp. 7,35,47-48). On the other hand, this approach is contrary to the practice of the Paninian Sanskrit grammatical tradition, which, in today’s terminology, starts with dependency relations and abstract features (§ 1.3) and uses a set of morphophonological rewriting rules to produce the corresponding surface forms (Kiparsky 2009).

The Classical Chinese grammatical tradition appears even looser compared with the Roman tradition in that the former did not even attempt to recognize parts of speech; this however was deeply rooted in the structure of Classical Chinese in that

Box 1.7: Ancient Chinese grammatical tradition and Roman tradition

Is the situation somehow close to what a Roman grammarian (*grammaticus*) would do?
It seems that Roman grammarians also didn’t care about abstract structures. See:

- Use and Function of Grammatical Examples in Roman Grammarians
- Quintilian’s ‘Grammar’ (Inst.1.4-8) and its Importance for the History of Roman Grammar
- What Graeco-Roman Grammar was about

On the other hand, phonology was an active topic in ancient China. This was probably due to influence of Buddhism: the need of translating Buddhist terminology into Chinese motivated systematic studies on Chinese phonology, and clear Sanskrit influences can be found in traditional works on phonology.

1.4.2 Perspectives of European missionaries

Large-scale, systematic and reliable grammatical description of Classical Chinese had unfortunately been largely lacking for quite a while, and until recently, when Classical Chinese was studied linguistically, it was the historical phonology that was studied (Pulleyblank 1995, p. xiii). Several missionary grammars however were still produced

Box 1.8: List of notable works

马若瑟汉语札记

1652年，卫匡国完成了《中国文法》一书。他将汉语划分出十大词类，并在动词部分对汉语主动语态和被动语态有所介绍。1703年万济国的《华语官话语法》，依旧沿袭拉丁语法，进行词类划分，并对其形态和范畴进行描写。除此之外，对汉语的一些造句规则（主要是主动句型和被动句型）也进行了归纳说明。1728年，马若瑟的《汉语札记》首次区分了汉语口语和书面语的语法规则，指出了汉语中“实字”与“虚字”的区别。不过马若瑟并没有依照拉丁语法构造汉语语法规则，而是从语言事实出发对汉语进行描写，这样的做法更符合汉语言的实际情况。

进入19世纪，随着欧洲汉学的兴起，越来越多的传教士和专业学者投身于汉语的语法研究中。法国汉学家雷慕莎的《汉文启蒙》在词类划分的基础上，探讨了一些句法规则，并指出汉语的语法不能用西方传统的拉丁语法去生搬硬套。在传教士方面，马礼逊的《通用汉言之法》其系统性比之前的语法专著更强，但主要内容还是集中在词法上，句法部分只归纳了九条规则，仅占五页篇幅。马士曼的《中国言法》在汉语语法本体研究上，也是由八大词类入手着重讨论词法，对句法的讨论较为简单，甚至有时句法与词法的界限并不清晰，将许多构词法放入了句法部分。艾约瑟的《中国官话语法》在前人基础上有所进步。除词法外，他对汉语句法规则的考察也颇为详细，这是此前西方人研究中所少见的。

1.4.3 Modern descriptions

Modern linguistic description of Classical Chinese has a clear starting point: the publication of 《马氏文通》 ‘Ma’s ’ by Ma Jianzhong (馬建忠).

TODO: 中等国文典, American structuralism in China for the study of Mandarin, Any other grammars in English besides Pulleyblank?

1.5 Remarkable features

Classical Chinese has several notable typological features.

Box 1.9: Remarkable features

- Wordhood
- Part of speech
- Topic-comment
- “Coverb”, or is there real preposition
- Prosody (and hence a chapter on phonology and writing system)
- The chapter on phonology and writing system can be very hard: lots of historical facts
- Passivization and so on
- Higher tolerance of ad-hoc recategorization: 名作动, 使动意动, etc.

Chapter 2

Grammatical overview

In this chapter we summarize the syntax of the clause (§ 2.2) and the syntax of the noun phrase (§ 2.3), along two intertwined lines: systems and constructions in clauses and noun phrases and grammatical functions therein, and lexical properties that determine how content roots appear in these environments.

Thus we identify nominal and verbal *grammatical environments* (i.e. the head of a noun phrase and the main verb of a clause), and identify noun and verb *form classes* in the lexicon according to language-specific properties of Classical Chinese at the same time, totally aware that the mapping between the two may be non-trivial (e.g. § 2.2.2.1).

2.1 Morphological typology

Grammatical wordhood has been a controversial topic in linguistics. Because Classical Chinese has no inflectional morphology, it is not easy to demarcate word boundaries. Indirect evidence from phonology, like the criterion of stress domain (Zhu 2009, pp. 41-42), is obviously not available for a dead language with little phonological recording.

Whether a composite form is a word cannot be easily seen from its structure building process, which often resembles phrasal syntax in modern Sinitic languages (Zhu 2009, p. 41).¹ It is hard to determine if such word-internal processes were already there during the Classical period.

One strongly indication of wordhood in Classical Chinese is AABB reduplication: phrasal or syntactic reduplication, already rare in comparison to morphological reduplication, tends to produce ABAB forms. These forms therefore are likely morphological reduplication (Travis 2001).

Another largely uncontroversial criterion of wordhood is lexicalization: an idiom (like *kick the bucket*) also has a single, coherent meaning, but it is much less common for a phrasal idiom to also impose strong constraints to its complementation and mod-

¹Zhu (2009, pp. 41-43) gives three other criteria besides the criterion of stress: (a) only words contain bound forms, (b) branches of a composite word cannot be modified, and (c) composite words have unpredictable meanings. As is admitted by Zhu, phenomena related to criterion (c) can be explained by idiomization. Criterion (a) is circular as it is repeating the definition of bound form: a verb obligatorily needs an argument to complement, but it does not make it a bound form. Criterion (b) phenomena can be simply explained by assuming that a *different kind of modification* is needed: consider *[ethno-[methodology]]-ical*, which structurally resembles phrases with modification. Without inflectional or phonological criteria, it is hard to explain how this form is fundamentally different from prototypical noun phrases.

ification. Thus the Mandarin verb 关心, despite having a regular verb-object structure, is a word, as it being transitive cannot be regularly inferred and thus has to involve lexicalization. The internal makeup of clauses and noun phrases depends on lexical properties of the head verb/noun, and the heads are grammatical words.

2.2 Clause

Like all other languages, a Classical Chinese clause can be a simple clause or a complex one constructed from subordination (§ 2.2.7) and coordination (§ 2.2.8).

A simple Classical Chinese clause is a nucleus clause plus discourse-related devices, including its speech act marked by sentence final particles (§ 2.2.6), and topicalization or focalization (§ 2.2.5). Topicalization can also happen for a complex clause (§ 2.2.8.1). These discourse-related devices typically only appear in sentential clauses and are absent in embedded clauses (except for direct quotations; § 2.2.2.7).

A nucleus clause may have nominal (§ 2.2.1) or verbal predication (§ 2.2.2). We call it a *nominal clause*² or a *verb clause*, respectively. Both constructions seem to have a well-defined subject position (which is not the same as topic), which however is often left blank (§ 2.2.4). Classical Chinese lacks obligatory tense, aspect, mood (TAM) marking but does have a TAM system marked by adverbs and possibly some sentence final particles (§ 2.2.3).

2.2.1 Nominal predication

The top-level structure of a Classical Chinese clause may contain an optional subject and a NP acting as the predicate (1, 2). A nominal predicate typically expresses an “is-a” relation between the subject (§ 2.2.4.3) and the predicate (1). Some nominal clauses however express a possessive relation between the two (2).

- (1) [此 人]_{subject} [力士]_{predicate}
this person strength-man
‘This person is a man of great strength.’ (史记·魏公子列传)
- (2) [蟹]_{subject} [六 跪 而 二 螯]_{predicate}
crab six leg CONJ two claw
‘A crab has six legs and two claws.’

We note that the nominal predication is nominal in the structural sense: a verbal clause can have a predicate headed by a form only recorded as a noun in the lexicon (cf. § 2.2.2.1).

Negation in Classical Chinese nominal clauses is usually expressed by 非, placed before the nominal predicate (3).

- (3) 凡群臣之言事秦者，皆奸人，非忠臣也

The nominal predicate is compatible with TAM adverbs (4).

- (4) 且 蔺相如 [素]_{TAM} 贱人
and NAME always humble-person
‘...and Lin Xiangru has always been a person with a very low social status.’

²We do not use the term to refer to a nominalized clause.

2.2.1.1 Copula constructions

All the constructions mentioned above are without a copula. In the pre-Classical copula age there is a copula 惟, which however had largely died out of use in Classical texts. Meanwhile, grammaticalization had added several copulas to Classical Chinese (Pulleyblank 1995, pp. 20-22).

2.2.2 Verbal predication

In describing clausal arguments, it is wise to separate (deep) slots in argument structure and (surface) grammatical relations. In morphologically richer languages, the mapping from these two to surface morphological cases may also be heterogeneous (Maling 2001); Classical Chinese has no case system and does not have quirky subjects marked by adpositions.

We use the standard S, A, P, etc. labels to refer to argument slots, without any implication of unity (S roles famously split into A-like and O-like ones, for instance).

The mapping from the argument slots to grammatical functions and surface morphological case marking is known as alignment. Classical Chinese has nominative-accusative alignment and hence has a well-defined subject: a clause with verbal predication consists of a *subject* (§ 2.2.4) and a verb phrase (VP), which is the predicate of the clause and contains TAM adverbs and core and peripheral arguments. Verb phrases can be coordinated (§ 2.2.8), demonstrating their constituent status. The subject, being outside the verb phrase, is sometimes known as the external argument in the literature (Huddleston and Pullum 2002, p. 53).

On the other hand, it is cross-linguistically harder to establish the *object* as a recurring grammatical function across different clausal constructions (cf. Pinkster 2015, pp. 28-29). The term *object* is used to refer to any argument without an explicit preposition for convenience (Li 2004, p. 102).

When TAM adverbs and peripheral arguments both appear before the verb, the order is always TAM > peripheral argument (5). The reverse order is never attested. The verb phrase therefore can be analyzed as the (extended) argument structure plus TAM adverbs preceding the pre-verbal peripheral arguments.

- (5) [三王]_{subject} [[既]_{TAM} [以 一]_{instrument} [定]_{main verb} [法度]_{object}]_{predicate:VP}
three-king already using determine law-system

‘The three great kings had already established social institutions by [it].’

The argument/adjunct distinction, or in the terminology of Dixon (2009), the distinction between core and peripheral arguments, is however much less clear. Phenomena commonly attributed to the distinction (like obligatoriness) can often simply be explained by semantic and pragmatic reasons and/or idiomization. Diagnostics based on structural closeness of the arguments to the main verb (e.g. pro-form substitution, extraction constraints) often fail to show qualitative distinctions. Lexicalization of the argument structure, arguably the golden standard of the distinction, is demonstrable in some but not all situations (McInnerney 2022, pp. 112-117, 125, 128, 212).

Lexicalization of verb frame is indeed attested in Classical Chinese (§ 2.2.2.2). The structural closeness of arguments to the main verb is briefly discussed in § 2.2.2.4.

2.2.2.1 The structure of the verb

Classical Chinese completely lacks prototypical verb inflectional morphology.

A peculiarity of Classical Chinese is that a root typically appearing in non-verbal environments can undergo ad hoc categorization (traditionally known as 词类活用) and serve as the main verb (or more accurately, the *predicator* i.e. the core of the predicate). (6) is a perfectly well-formed verbal clause: it expresses a dynamic event and not just a state and has a modality auxiliary (illustrating that the predicate is not a noun phrase). (8) is a stative verbal clause involving ad hoc categorization.

- (6) [假 舟 楫 者]_{subject}, [非 能 水 也]_{predicate:VP}
draw.help boat paddle REL NEG can swim SFP
'Those who draw help from boats and paddles cannot swim, ...'

This ad hoc categorization process differs from prototypical derivation in that it is irregular (and thus it is not possible to generalize from the pattern in 6 to a generic process), and therefore in the terminology of Huddleston and Pullum (2002, pp. 1640-1641) is conversion as opposed to derivation. Unlike the case in formal English,³ in Classical Chinese conversion to forms not lexically well established is idiomatic: that 水 appears in (6) does not necessarily mean creation of a new verb *lexical entry*. In traditional terms, 词类活用 does not mean 一词多义.

In other clauses, the meaning of the root filling the main verb position of the clause can be regularly inferred, and should thus be considered verbalization derivation. An important process of this kind is the tropative (35).

Whether certain multimorphemic forms should be considered outputs of verb derivation is not completely clear, because of the blurry boundary between words and larger units (§ 2.1). As an instance, it is hard to determine if the second verb is incorporated into the first verb in the sequence 问曰 (which comes from argument structure-level coordination; cf. 19).

2.2.2.2 Intransitive and transitive constructions

The constituent order of transitive clauses is almost always SVO (7; but see § 2.2.2.9). Intransitive clauses have a SV constituent order (8).

- (7) [子]_{subject,A} [[奚]_{reason (§ 2.2.2.9)} 不 [为]_{verb,DO} [政]_P]_{predicate:VP} ?
(8) [君子]_S [不 [器]_{verb,BE}]_{predicate:VP}
lord-son NEG instrument

'A gentleman does not resemble an instrument/container (i.e. his skills are broad).'

Consistent with cross-linguistically attested patterns, in Classical Chinese, a mono-transitive or intransitive clause can be about an intentionally initiated event (DO; 7, § 8.1.1), a state (BE; 8, § 8.1.2) or a change of the state (BECOME; § 8.1.2). The DO type can further be divided into the transitive and intransitive classes. BE and BECOME clauses are intransitive by definition. Finally, BECOME and BE clauses are often inputs to the causative construction, resulting in CAUSE-BECOME/BE clauses.

³Ad hoc conversion however is possible in informal English, as in *I might [guinea pig] it for you* and even the dephrasal "*Sappho and her friend*"d.

The distinction between these constructions has long been noticed (Li 2004, pp. 92-96)⁴ and has consequences for animacy and volition of the subject as well as the viability of argument structure alternation processes: S=A labile valency alternation naturally exists for DO clauses (§ 8.2.1.1), and S=P labile valency alternation naturally exists for CAUSE clauses (§ 8.2.2.1).

Whether a verb is able to head a DO, BE, BECOME or CAUSE clause is lexically specified and is an important parameter in subclassifying verbs: for instance, some causative verbs have no intransitive counterparts (§ 8.2.2.2) and eventually are reanalyzed as a DO verb (Mei 2015, pp. 269-271), and not all DO verbs have S=A labile valency (Li 2004, p. 93).

Some verbs also allow quantitative objects (33), and are “transitive” superficially with a structure different from that of (7).

2.2.2.3 Passive and causative constructions

Classical Chinese has a regular causative valency increasing construction producing CAUSE clauses (§ 9.1). It typically is not compatible with a verb frame that already has an A-like argument (Mei 2015, p. 364).

There are also productive valency decreasing constructions which may be known as the passive, which suppress the subject and promote an internal argument to the subject position (§ 9.2). These constructions are different from prototypical passive *voice* constructions in that they are much more selective about the original verb frame.

(9) is an example of the causative applied on top of the passive.

- (9) 杀 御叔
kill NAME
‘...caused Yushu to be killed.’ (=1 in § 9.1.1)

Classical Chinese seems to already have a prototype of what is later known as the disposal construction or the 把 construction in later Sinitic languages. The sequence of 以 plus the argument following it is never fronted, implying that the construction is not a prepositional argument construction, but a light verb construction. Other light verb constructions include the 为...所 construction and more.

- (10) 尽以其宝器赂献于周厘王

2.2.2.4 Prepositional arguments

(11, 12, 13) are examples of prepositional constituents, carrying a wide range of semantic roles. When an object is present, a rigid prepositional argument-after-object linear order can be observed after the main verb (12, 13). Prepositional phrases can also be fronted (§ 2.2.2.9).

- (11) 公 惧， 队 [于 车]_{source}
king afraid fall from carriage
‘The king was afraid and fell from the carriage.’

⁴More accurately, Li (2004) classifies subtypes of objects (受动宾语 vs. 致动宾语). His classification maps exactly to the DO vs. CAUSE-BECOME distinction. He also emphasizes what is essentially the difference between the S=A DO labile valency and the S=P BECOME labile valency.

- (12) 故 賜 [之] [以 重 祭]_{theme}
 therefore give 3 PREP solemn offering
 ‘...and therefore gave him solemn veneration.’
- (13) 季康子 問 政 [于 孔子]_{location}
 NAME ask politics PREP NAME
 ‘J. asks Confucius about politics.’

One issue surrounding licensing of prepositional arguments is whether some of them are licensed in so-called small clauses (Punske 2013). No convincing evidence for a small clause analysis is found in Classical Chinese; clauses with structures comparable to (13) cannot be analyzed in the small clause framework anyway (Mei 2015, pp. 417-419). Regardless of their licensing, prepositional arguments stay in situ in every type of valency alternation, except for the applicative. Many preposition arguments can be safely viewed as the least pivotal and the most internal argument (Mei 2015, p. 404), though circumstantial constituents, like those in the 見...於 construction, are likely structurally more external (Mei 2015, p. 323).

Some prepositional arguments can be replaced by non-prepositional objects (14). This is not possible for every verb: no non-prepositional counterpart of (11) is attested.

- (14) 入秦何見

2.2.2.5 Applicative constructions

Besides post-verbal preposition omission, Classical Chinese has applicative constructions that turn a clause containing a prepositional argument into a double object clause (Mei 2015, Ch. 10).

(15) illustrates an applicative clause where the target of the question becomes an object (cf. 13). The prepositional argument appears to be promoted to a more pivotal position (§ 2.2.4.2), as demonstrated by the linear order.

- (15) 上問上林尉諸禽獸簿

Other structurally comparable constructions include the tropative and the benefactive (Mei 2015, pp. 389-415). Mei (2015, p. 420-421) also notes that ditransitive clauses of giving and receiving in Classical Chinese may have a similar structure.

Not all verb frames have applicative counterparts. T-before-G applicative counterparts of (12) for instance are not attested. Double objects may still appear by removing the preposition (cf. 14; § 8.3.1).

2.2.2.6 Possession in argument structure

External possession is a heterogeneous phenomenon attested cross-linguistically (Deal 2017). Classical Chinese has a subject external possession construction, apparently structurally parallel to Japanese external possession (16).

- (16) 子路性鄙

(16) should not be analyzed as topicalization, because typically a pause cannot be inserted after 子路 (§ 2.2.4.1).

2.2.2.7 Complement clause constructions

Besides raising and control, Classical Chinese has at least two complement clause constructions. It is not possible for the prototypical complement clause in (17) to have sentence final particles. On the other hand, the direct quotation in (18) is sentential and has a sentence final particle.

(17) 臣窃以为 [不便於君]_{complement clause}

(18) 子曰：[“不患人之不己知，患不知人也。”]_{direct quotation}

Whether the complement clause is a direct quotation is lexically determined. In Classical Chinese, direct quotation is taken by a handful of verbs, the most frequent one being 曰.

Other verbs of speech, which themselves do not license complement clauses (20), can introduce direct quotation by argument structure-level coordination (§ 2.2.8; 19).

(19) [子禽]_{subject: NP} [[问]_{verb} [于 子贡]_{target}]_{verb phrase 1}
NAME ask at NAME
[[曰: “...”]_{direct quotation}]_{verb phrase 2}
say

‘Ziqin asked Zigong: ...’

(20) 季康子患盜，問於孔子。

2.2.2.8 Raising and control

It is possible for an argument of the complement clause (usually the A argument; § 2.2.4.2) to appear as a clausal dependent in the main clause. Depending on when that argument is a part of the deep argument structure of the main clause or not, such an argument-sharing construction can be classified as control or raising. The underlying mechanisms of control constructions may again be heterogeneous, a topic beyond the scope of this work.

(21) is an example of raising-to-object in Classical Chinese, and is also an example of lexical causative (§ 9.1.2). This construction is sometimes known as the *pivot construction* (Pulleyblank 1995, p. 40) or 兼语式 in Chinese (Mei 2015, p. 375),

(21) 令军勿敢犯

Uncontroversial pivot constructions are limited in number in early texts, because many of them can also be analyzed as argument structure-level coordination (Mei 2015, p. 376; § 2.2.8).

Control can also be observed in Classical Chinese.

(22) 吾不欲觀之矣

2.2.2.9 Verb phrase structure

All arguments (He 2005, pp. 3-5), include prepositional ones (59), can be omitted when they are identifiable from the context.

Certain verb phrase constituents can appear before the main verb. TAM adverbs almost appear before the main verb. Below we discuss fronting of (core and peripheral) arguments.

First, the object pronoun is often regularly moved before the verb in a negative clause (23) without clear information structure motivation, leading to a SOV order. An interrogative object pronoun can also be fronted (24). This may be analyzed as the pronoun clitic being attached to the negator or the main verb (Aldridge 2010).

- (23) 余 [忍]_{predicate}
 1 NEG 2 endure kill
 ‘I can’t bear to kill you.’ (左传)
- (24) 以五十步笑百步，则 [何如]_{SOV interrogative clause}
- (25) 我未之能易也

Second, Classical Chinese appears to have a verb phrase internal focus position (Mei 2015, pp. 323-324; cf. Danckaert 2011).

In (26), the pre-verbal constituent 以其妹 seems to be a focalized argument. It cannot be the sentential focus, because the subject 季康子 seems to stay in-situ with no pause after it and therefore cannot be the sentential topic.

- (26) 季康子 [以其妹]_{VP-focus_i} 妻之 -_i

The fronted prepositional phrase may further undergo preposition-object inversion (27). This is much rarer than fronting of object pronouns, which again is consistent with verb phrase internal focalization.

- (27) 室於怒而市於色

Interestingly, object fronting is usually accompanied by presence of 是 or 之 before the main verb (TODO: ref).

2.2.3 Tense, aspect, and modality

Syntactic TAM categories (not just devices with semantic TAM meaning) are within a single clause and form a system with a hierarchy of auxiliaries and adverbs. Thus English *should have been doing* has grammaticalized TAM marking by auxiliaries, as the relative order of the auxiliaries cannot be shifted even with semantic motivations. Adverbs like *now* or *frankly* follow a similar order constraint and are a part of the TAM system (Cinque 1999).⁵

Because Classical Chinese have no native speakers, it is not possible to know if certain forms are impossible, and therefore there is always room to argue that an alleged auxiliary verb is in fact a content verb or certain unattested word orders are in fact possible. Further, some temporal clausal modifiers may be circumstantial peripheral arguments and not TAM adverbs (Cinque 1999, pp. 29-30; Mei 2015, pp. 430-431). The analyses below therefore are all tentative to a certain degree.

⁵On the other hand, *be able to* is not a grammaticalized TAM marker (yet), as *many of us would have been able to have made this kind of bomb* is attested, where the PERFECT marking appears both before and after *able*, modifying both the fact many of us being able to do something and the thing being done (i.e. making this kind of bomb), strongly suggesting that we are looking at a biclausal construction.

2.2.3.1 Speaker-oriented categories

The most wide-scope TAM categories are usually categories related to the speaker's evaluation of the situation being talked about. TAM categories falling under this type include evaluative adverbs (like *frankly* at the start of a sentence), evidentiality, and epistemic modality.

Classical Chinese has a category somewhere between evidentiality and epistemic modality, which may be known as speaker's commitment and has a subjective/objective distinction. Strong speaker's commitment to objectivity of the state of affair being reported can be marked by the adverb 固 (28). That the adverb is before the anteriority adverb 已 demonstrates its high-scope, speaker-oriented nature.

(28) [固] [已] 多敗亡

On the other hand, Li (2004, p. 359) notes that clauses with certain sentence final particles, which he calls 说明句, are more subjective. The sentence initial particle 盖 also marks an inferential attitude towards the situation being reported.

Some speaker-oriented adverbs, like 殆 in (30), may be interpreted as epistemic modality.

Classical Chinese has one category known as the *sentential aspect*, marked by the sentence final particle 矣 (29). Its function is close to the sentence final 了 in Modern Standard Mandarin (Paul 2014b; Pan 2021): it is a wide-scope category indicating that the reported situation of affair is new (Pulleyblank 1995, p. 116), and has been compared with inchoativity/evidentiality (Tsai 2008) and is comparable to the iative or the new situation marker in Southeastern Asian languages (Mazzitelli 2025).

(29) 此迫矣

2.2.3.2 Tense and modality

Tense adverbs are attested in Classical Chinese. (30) is an instance of adverbial marking of future.

(30) 夫子 [殆]_{epistemic} [將]_{future} 病也

Mei (2015, p. 428) proposes that the sentence final particle 矣 introduces an explicit deictic time and therefore should be considered a (weak) tense marker. Pulleyblank (1995, p. 116), on the other hand, analyzes the particle as a “new information” sentential aspect.

Non-epistemic wide-scope modality categories about whether something has a chance to happen exist in Classical Chinese. (31) illustrates a possibility adverb. A realis/irrealis distinction can be found in conditional constructions (51), relevant to the licensing of TAM markers (Mei 2015, p. 81).

(31) 我 [固]_{objectivity} [能]_{possibility} 治偏枯

2.2.3.3 Aspect

What is known as aspect includes anteriority i.e. secondary tense (English *have done*; Huddleston and Pullum 2002, p. 140), (Rizzi and Cinque 2016), (im)perfectivity i.e. point of view aspect or progressive aspect, the habitual aspect, and a lot more. These categories often have scope comparable to that of volition (Rizzi and Cinque 2016).

We have already seen an instance of anteriority (28). 方 in (32) is a progressive adverb, demonstrating existence of point of view aspect in Classical Chinese. The adverb also has a prospective meaning.

(32) 秦王方環柱走

2.2.3.4 Lexical aspect

The lexical aspect is correlated to both argument structure and TAM categories (Meisterernst 2013; Meisterernst 2016). (33) illustrates how presence of a quantity argument turns the clause telic. (34) illustrates how the habitual aspect coerces an atelic reading. It has also been observed that stative verbs are not compatible with perfective adverbs (Meisterernst 2016).

(33) 行二百餘里

(34) [常]_{habitual} 封三錢之府

2.2.4 Subjecthood

In § 2.2.1 and in § 2.2.2, we both mention the concept of *subject*. This concept of a single, unitary pivot in a clause should not be taken for granted.

Two types of *pivots* exist in a clause: one based on argument structure properties like obligatory argument omission in control constructions or binding of reflexive pronouns (§ 2.2.4.2), another based on clausal pivot properties like subject sharing in coordination and relativization (§ 2.2.4.3). Prototypical subjecthood is identification of the two pivots and hence absence of syntactic ergativity (Aldridge 2008), discussed in the sections below.

Another problem is the distinction between subject and topic, both appearing at the beginning of a clause with unclear structural differences. Some scholars treat any seemingly pivotal constituent a subject (e.g. Li 2004, p. 41) and hence the topic is a “big subject” (Li 2004, p. 42). This opinion appears to suggest that Classical Chinese has only information structure and no argument structure in its syntax (Mei 2015, p. 122). We discuss the issue in § 2.2.4.1.

2.2.4.1 Subject vs. topic

We argue that identifying subject and topic (as in Li 2004, pp. 41-42) is not desirable for describing Classical Chinese.

First, in reading traditions of Classical Chinese (as indicated by traditional 句读 punctuations), some pivotal constituents are followed by a pause, while others are obligatorily not followed by a pause (Mei 2015, pp. 121-122). No existing theory equating subject and topic naturally explains the distinction; on the other hand, availability of the pause is a cross-linguistic indication of topicalization.

Second, in trophative clauses, the pivotal constituent closest to the main verb obligatorily refers to the person holding the opinion. (35), where the main verb is obviously a trophative use of the noun 客 ‘guest’, can only be interpreted as ‘Lord Mengchang considers me as a guest’ (and never ‘I consider Lord Mengchang as a guest’). This implies that the argument structure of the clause imposes a hard constraint on what argument can be promoted to a pivotal constituent, which is unusual for topicalization.

(35) 孟尝君客我

When the pre-verbal pivotal constituent closest to the main verb is semantically the patient, it is often the case that it is obligatorily so (36), which again is unusual for topicalization (but not unusual if interpreted as valency alternation).

(36) 国定而天下定

Therefore, pivotal constituents that are not topics and are structurally closer to the main verb than prototypical topics exist in Classical Chinese.

2.2.4.2 Pivot in argument structure

The existence of a pivot at the level of argument structure, which is often the A-like argument in a clause, can be demonstrated by examining the shared argument in control constructions. The reflexive pronoun binding test is not available in Classical Chinese, as no unambiguous reflexive is attested.

Further, the existence of an argument structure pivot can be confirmed by observing valency alternation constructions, as most of them involve either promotion of an argument to an A-like position or removal of the A-like argument.

It is sometimes also possible to define the second most pivotal argument. In applicative clauses, the promoted argument undergoes a leftward movement, which can be interpreted as it being promoted to a structurally more pivotal position (§ 2.2.2.5, 15). In double object ‘giving’ clauses, it is always the G argument being passivized (37), which again can be interpreted as the G argument being structurally more pivotal.

(37) [诸侯]_G, 赐 [弓矢]_T 然后征

Hierarchies of pivotality or “externality” of arguments are not uncommon in world languages.⁶

2.2.4.3 Clausal pivot

We then proceed to prove that the A-like, most pivotal argument in a verbal clause and the initial constituent in a nominal clause are clause pivots, around which everything else “revolves”. This proves that Classical Chinese has nominative-accusative alignment and justifies the notion of subject.

The clearest criterion that defines clausal pivotal subjecthood is probably coordination: when clauses are coordinated, one constituent seems to be shared by all of them, then this constituent is probably the clausal pivot, which is usually the argument structure pivot (§ 2.2.8, 52).

Note that two coordinated clauses can also share a topic, but there are signs which tell us that what is shared is the topic and not the subject (e.g. § 2.2.8, 53, where the topic is the object of the first clause and the subject of the second clause).

⁶In Japhug, a language with direct-inverse alignment, it is possible to have a clause containing a causer, an instrument, an agent and a patient. The personal indexation marker appears to be decided on the basis of the hierarchy *causer*→*instrument*→*agent*→*patient*, by first taking the two most internal arguments and deciding which is more salient on the empathy hierarchy, then comparing the result with the third most internal argument and deciding which is more salient, and finally comparing the result of the last step with the most external argument; hence 1→3→2→3 is morphologically equivalent to 1→2 (Jacques 2021, p. 310, p. 584, (116), p. 848, (67)). Similar mechanisms are attested cross-linguistically (Wiltschko (2014, § 7.4.1)).

2.2.5 Information packaging

2.2.5.1 Topicalization

Topicalization is ubiquitous in Classical Chinese. Indeed, the vanilla nominal predication construction (1) is much less frequent than the so-called 判断句 ‘judgmental clause’, in which the subject is topicalized, often with the sentence final particle 也 (38), or the topic particle 者 (39), or both.

- (38) [城北 徐-公]_{topic: NP}, [齐-国 之 美-丽 者]_{comment: § 2.3.4}
city north NAME-GONG Qi-country GEN beautiful-beautiful REL
也
SFP
‘Mr. Xu from the north of the city is a handsome guy in the country Qi.’
- (39) [兵]_{topic: NP} 者, [不 祥 之 器]_{comment, predicate}
weapon TOPIC NEG fortunate GEN instrument
‘Weapons are not auspicious.’

The topic marker 者 is not restricted to the nominal predication construction (40).

- (40) 此二人者, 实弑寡君

What is topicalized is of course not restricted to the subject. This fact is a piece of evidence supporting the distinction between subject and topic in Classical Chinese. In (41), the comment clearly has a nominal predicate. What is promoted to the topic position however is not the subject of the nominal predicate construction, but the predicate. The subject is likely the *focus* and not the topic (Mei 2015, p. 138). Topicalization of other clausal constituents is also possible (e.g. § 2.2.8, 53).

- (41) [仁之实]_{topic:NP_i}, [[事亲]_{subject} [是]_{predicate_i}]_{comment} 也

We also note that the 者…也… framework is not limited to topicalization of nominal predicate clauses (i.e. “judgmental clauses”). For instance, we have (42), in which the sole argument in an existential clause is topicalized, and the comment receives 也 as its sentence final particle.

- (42) 然而不王者, 未之有也

We also note that topicalization can happen multiple times (43).

- (43) 万乘之国, 弑其君者, 必千乘之家

Some topic-like constituents are analyzed as *frames* in Mei (2015, p. 129).

2.2.5.2 Focalization

Topicalization is marked by fronting and a pause, but what is fronted and before a pause is not necessarily a topic. In (44), for instance, the verbal predicate is fronted, which likely is not topical. Note that the sentence final particle is fronted as well.

Box 2.1: Fronted SFP

The phenomenon can be analyzed in multiple ways. We may assume that (44) is essentially some sort of cleft construction, in which the subject is first separated from the rest of the clause and then the rest of the clause is focalized. Or we can analyze 矣 as a TAM marker, and not a marker from the CP layer. Or maybe we can argue that markers from CP layers are morphologically verbal and have to be attached to either the main verb or the verb phrase. Which analysis works best depends on whether they are consistent with other phenomena.

- Can we prove that the SFPs are very “high-level” and are above the topic layer? For example, can two clauses with different subjects share a SFP?
- Semantically do 矣 carry TAM meanings?

- (44) [[甚]_{VP_i} 矣]_{focus}, [[汝 之 不 惠]_{subject:clause (§ 2.2.6.2) —predicate_i}]_{nucleus clause}
 extreme SFP 2 GEN NEG smart
 ‘You are so stupid! (lit. So extreme is your being unintelligent!)’

Box 2.2: A complete overview of the left periphery

See <https://referenceworks.brill.com/display/entries/ECLO/COM-000248.xml>

2.2.6 Speech acts and clause types

This section discusses high-level grammatical categories which, in the case of sentential clauses, are often related to the speech act of the clause. These categories are traditionally known as *clause types* or 句类, as opposed to 句型 i.e. the internal structure of clauses (Li 2004, p. 58).

2.2.6.1 Sentential clauses: interrogative, exclamative, and imperative

The speech force of sentential clauses are marked by sentence final particles. In fact, these particles almost only appear in sentential clauses.

Interrogative clauses may completely lack formal marking except interrogative pro-forms (24). The interrogative speech force is optionally marked by 乎 and other particles (45). The exclamative speech act is similarly marked by sentence final particles (46).

(45) 其能久乎?

(46) 吾死矣夫!

We note that in (46), the sentential aspect marker 矣 (§ 2.2.3.1) appears before the exclamative 夫, which means that the two systems of particles can coexist. The reverse order *夫矣 is not attested.

2.2.6.2 Nonfinite constructions

Classical Chinese has one clausal construction that at the first glance resembles the noun phrase. Instances of it appear as the object (47) or as a subordinated clause, like a conditional or temporal clause (48; § 2.2.7).

The construction is sometimes known as nominalization (Pulleyblank 1995, p. 64). This term cannot be taken literally, because the construction does not admit adjectival modification, while there is nothing resembling an object (e.g. 子 in 48) in prototypical Classical Chinese noun phrases. The construction therefore has a structure comparable to the English gerundive non-finite clause *his playing national anthem* and is structurally different from the noun phrase.

- (47) 王 如 知 此, 则 无 望 [民 之 多 于 邻
king if know this then NEG hope people GEN more than neighbor
国]object: gerundive 也
country SFP
'If Your Majesty knows this, then don't expect your people to be more plentiful
than your neighboring countries' people.'
- (48) [父母之爱子]condition: gerundive, 则为之计深远

2.2.7 Subordination

The term *subordination* sometimes means all kinds of clause embedding. In this section we primarily focus on bipartite clauses with the structure and meaning of 'if ... then ...' or 'when ...'. Complement clauses are discussed in § 2.2.2.7 and § 2.2.2.8. Relative clauses are discussed in TODO

2.2.7.1 Sentential subordination

An overview of subordination constructions in Classical Chinese can be found in Mei (2015, Ch. 3). In all Classical Chinese conditional constructions, the condition usually appears before the consequence (49, 50, 51). The consequence can be marked by a particle (e.g. 则; 49). Sometimes the marker 则 is dropped (50) but putting it back should never render a sentence ungrammatical (Mei 2015, p. 86).

The condition clause can also be marked. Classical Chinese distinguishes between realis and irrealis conditional constructions (§ 2.2.3.2): the former are marked by e.g. 既 (49), while the latter are marked by e.g. 若 (51).

- (49) [既来之]condition, [则安之]consequence
(50) [杀女]condition, [我伐之]consequence
(51) [若已食] 则退

Box 2.3: Position of condition marker

When the subjects of the two clauses are shared, it seems 若 obligatorily appears after the subject of the first clause. A possible analysis is to assume that the subordination construction is working at the level of VPs.

An interesting phenomenon is that the subordinated clause can be a nonfinite 之 clause (48). This is cross-linguistically unsurprising (as in Japanese or Turkish). Note that the marker 若 can be attached to the gerundive condition clause as well (Mei 2015, p. 98).

2.2.8 Coordination

Coordination happens at the level of clause, (extended) verb phrase or argument structure. The distinction between the last two is that coordination of two argument structures results in a *single* situation of affairs, usually with shared TAM marking, while coordination of two verb phrases is more comparable to clausal coordination (Mei 2015, pp. 192-201). The two also have differences in relativization (Mei 2015, p. 207).

In Classical Chinese, certain examples are clearly argument structure-level coordination (19). Others are clearly not, including those in which the two verb phrase branches are separated by a pause (Mei 2015, p. 195). However, due to the absence of obligatory TAM marking (and unavailability of acceptability judgement data), marginal cases in the classification above exist in Classical Chinese. See (TODO) for a detailed analysis.

Coordination is primarily marked by 而 (52), which is only licensed in a clausal environment and cannot link to nominal constituents. Note that the functions of 而 are not restricted to conjunction (Mei 2015, p. 183).

(52) 声伯…… [食使者]_{verb phrase 1}, 而后 [食]_{verb phrase 2}

2.2.8.1 Topic chains as syntactic coordination

An interesting question is the interaction between topicalization and coordination. Mei (2015, p. 217) contends that “topic chains”, i.e. several clauses with a shared topic (Mei 2015, Ch. 4 § 3.3), are discourse structures and not syntactic structures. Therefore topicalization happens first, and coordination happens then: after that no further topicalization is possible. He further argues that clauses in a topic chain cannot be linked together by 而. (53) however seems to be a counterexample. This example clearly contains two coordinated clauses. In the first clause 取之于蓝, 之, appearing after the verb, can only be a pronoun, and the only sensible reading of the clause is that 之 (the object) is coreferential with 青 at the initial of the sentence, and 取之于蓝 then means ‘(people) extract it (i.e. indigo dye) from *Indigofera*.’ Therefore, 青 at the initial of the sentence is the object of the first clause and the subject of the second clause, meaning it cannot be the shared subject. This, together with the traditional pause after the first 青, means the first 青 likely is a topic, which means here topicalization happens *after* coordination.

(53) [青]_{topic: NP_i}, 一_i 取 [之]_{object: Pronoun_i} 于 蓝 而 一_i 青 于
indigo.dye pick it from *Indigofera* conj blue than
蓝
Indigofera
‘Indigo dye, people extract it from *Indigofera*, but it’s bluer than *Indigofera*.’

2.3 The noun phrase

The Classical Chinese NP can be roughly divided into the determiner region and the “core” region, the latter known in Huddleston and Pullum (2002) as the *nominal*.⁷ The

⁷In this note, when the term *nominal* is used as a noun, it refers to the determined region in NPs, while when it is used as an adjective, it refers to the status of being the head of a NP.

latter is the head noun plus possible arguments and modifiers, and the former can be left empty or filled by a demonstrative or a “possessor”, the role of the latter being not confined to a semantic possessor (Pulleyblank 1995, p. 61).

2.3.1 Structure of the noun

2.3.2 Argument structure

2.3.3 Direct modification adjectives

(54) 大白狗 (汉书·五行志)

However, examples involving multiple pre-nominal attributives are scarce. Complex attributives are much more idiomatically constructed using the relative clause construction (55).

2.3.4 Relative clause constructions

The marker 者 looks like a relativizer. It is different from relativizers in many other languages in that further structural add-ons can be applied to the fused relative clause formed by it, while the fused relative clause constructions in many other languages are unable to undergo further modification.

(55) 若[至力农畜，工虞商贾，为权利以成富，大者倾郡，中者倾县，下者倾乡里者]，不可胜数

Box 2.4: Relative clause complexity

Can a relative clause contain a NP that in turn contains a relative clause?

Box 2.5: zhi-zhe construction

The structure of the 之-者 construction may cause some debates. It can be analyzed as a possessive construction on top of a fused relative clause construction and translated word-to-word into English as ‘[those who go one thousand miles] of horses’. An interesting question then is whether we have any other instances of the N 之 V 者 construction where the relation between N and [V 者] is prototypically possessive. It seems this is indeed possible: 城北徐公，齐国之美丽者也。

Under this analysis, 楚人有吹箫于市者 is composed by applying the external possessive construction to 楚人之吹箫于市者

One fact (or is it really a fact?) supporting the determinative analysis of 之-者 is that the construction seems to be unable to receive a further determiner: *此马之千里者. The sequence 此马之千里者 does appear but it is almost always a nominal predication construction.

Box 2.6: What can be relativized, and possible external possession

若至[力农畜，工虞商贾，为权利以成富，大者倾郡，中者倾县，下者倾乡里者]，不可胜数

It seems what is relativized here is the subject of the bracketed clause. But then what’s the role of 大者倾郡? If we consider it to be a coordinated clause, then it seems an argument is moved from only one branch of a coordination construction: a clear

violation of the coordinate structure constraint of extraction!

If we consider it to be a coordinated VP, then Classical Chinese should have an external possession construction: [商人]_{subject} [大者倾郡]_{predicate}, in which 大者 is a part of 商人.

Or maybe this is a clausal pseudo-coordination: *what did Alex go to the store and buy*.

2.3.5 The determiner system

When the “possessor” is present, the particle 之 appears between the possessor and the nominal region (56, 57). When only the demonstrative is present, no marking is present (58).

(56) 王之诸臣

(57) 马之死者

(58) [此心] 之所以合于王者

2.3.6 Prepositions

In Old Chinese, there are only two prepositions: 于 and 於. The exact usages of the two prepositions are not clear. In *Zuo Zhuan*, 于 is reserved for prepositional complements (§ 2.2.2.4), while 於 is for inter-predicate focalization (TODO). Other Old Chinese works have different conventions.

It is possible to omit the object of a preposition.

(59) 孔子 [因 —]_{reason} 叹

2.4 Parts of speech division

Having had an overview of grammatical constructions in Classical Chinese, we turn to analyze the structure of its lexicon. That is to say, we now study the parts of speech division in Classical Chinese.

Classical Chinese has no inflectional morphology for content words, so it is not possible to define parts of speech based on inflections. Content words show much flexibility in their distribution in various syntactic environments, sometimes without any formal indications. These facts lead some to claim that Classical Chinese is a language without clear part of speech distinctions, so although we can talk about the nominal or verbal usage of a root or a compound, strictly speaking we cannot talk about nouns or verbs, as there are no inherent lexical properties attached to roots that dictate their nominal or verbal usages. A more careful analysis, though, seems to reveal that at least some part of speech distinctions can be maintained in Classical Chinese, although Classical Chinese is much more tolerant to ad hoc re-categorization of roots than, say, English.

In principle, function words can be introduced together with their grammatical functions, but since the correct analyses of some constructions are still controversial and it may well be possible that the controversies reflect real historical linguistic divergence among speakers, function words are also discussed here for easier reference.

2.4.1 Nouns and verbs

A noun-verb distinction is supported by carefully examining the traditionally recognized noun-used-as-verb phenomena (§ 7.1.1). If the lexicon of Classical Chinese contains *only* non-categorized roots, the interpretation of verbal usages of a word that usually appears in nominal environments should vary rather freely. What is actually attested however is not different from similar phenomena in other languages. In some cases, it seems a root is first categorized as a noun and then undergoes something similar to English *-ize* (albeit without any explicit marking), so only the nominal usage needs to be recorded as a lexical entry, but the lexicon controls whether a derivation step is viable. In other cases, both the nominal and verbal usages are to be recorded in the lexicon, as they cannot be inferred regularly from each other. In both cases, how a root is possibly categorized is stored in the lexicon, meaning that calling the nominal use of a root a *noun* and the verbal use of a root a *verb* is not problematic at all even in Classical Chinese. Sporadic ad hoc re-categorization of roots does exist, but this does not support the idea that part of speech division does not exist at all in the lexicon.

A terminological caveat is what appears as an argument is not necessarily a NP: it can be a complement clause. The main verb of a complement clause is not in a nominal position. Some may call complement clauses “nominal clauses”, but this is misleading as the internal structure of a complement clause is not the same as that of a NP.

2.4.2 The adjective class

An adjective class can also be established in Classical Chinese, although its behavior is strongly verbal.

A caveat, similar to the caveat that an argument is not necessarily a NP, is that an attributive phrase is not always an adjective phrase. In existing modern studies, statements like “a verb used as an adjective” is usually avoided: wordings like “something is used as an attributive” are adopted instead.

Box 2.7: Traditional grammars

List some Classical Chinese grammars in which 动词作形容词 etc. never appear.

Box 2.8: A comprehensive list of Classical Chinese parts of speech

Noun, verb, adjective: any other content words?

2.4.3 Particles

Grammatical particles are not content words and in principle can be introduced together with the grammatical categories and relations they express. The long and complicated historical evolution of Classical Chinese however means a particle may have multiple quite different uses possibly due to grammaticalization, so a surface form-to-function discussion on particles is of great descriptive value.

Box 2.9: Classification of particles

Do I need to classify particles?

者 The particle 者 most frequently appears as a relativizer, a complementizer, or in the *zhe-ye* construction. The three functions can be uniformly analyzed as the function of a low-level determiner (Aldridge 2009).

之 (TODO: add a discussion of the particle 之.)

Chapter 3

Phonology and the writing system

Box 3.1: On the writing system

- 隶定和简化: one keeps the structure of a character and only alters the components, another messes up the structure
- 谐声
- Syntax within the character?

3.1 Theoretical consequences

Now we discuss the cognitive status of the “grammar of characters” sketched above. Questions relevant to this topic include whether grammar-like rules governing the structure of Chinese characters are synchronic or historical, and if they are synchronic, whether they derive from human language capacity or from some other cognitive capacities. We note that the latter question is ultimately related to the big questions in theoretical linguistics and cognitive science (see also discussions at the end of § 1.3): if the grammar of Chinese characters mimics the grammar of spoken natural languages but the network in the brain processing Chinese characters is completely independent of the language network, then what we thought were unique to languages probably are not domain-specific to languages at all.

Regarding the first question, neurolinguistic experiments suggest that both holistic and sub-lexical processing exist in the human brain. The existence of holistic processing is supported by the fact that Chinese readers find it easier to tell completely different characters apart than to tell characters with shared components apart, while the existence of sub-lexical processing is supported by the fact that characters with valid semantic or phonetic components are processed more quickly (Duan and Cai 2024, § 2.2). So literate Chinese speakers do have the components (subconsciously) in mind when reading Chinese characters. Yet the same can be said for all orthographic systems (Myers 2019, pp. 23-25). English orthography, for instance, has phonology-like rules like $-y + \text{suffix} \rightarrow -i\text{-suffix}$ (Myers 2019, p. 26).

What makes Chinese characters special is whether they have non-regular structures.

Regarding the neurolinguistic properties of Chinese characters, we note that the brain region primarily responsible for recognizing Chinese characters is the Visual Word Form Area, which is not a part of the language network in the brain, and this suffices as an argument against the assumption that the “grammar” of Chinese characters and the grammars of spoken languages have the same neurological origin (Myers 2019, pp. 209-210). Still, this does not by itself settle the cognitive status of character-internal structure.

Chapter 4

Nouns

4.1 Derivation

The verbs 出 ('go out'), 入 ('enter'), 亡 ('die, decay') are regularly derived to 出 ('what goes out'), 入 ('what comes in') and 亡 ('what dies').

Box 4.1: Deverbalization derivation

Summarize deverbal derivations.

4.2 Proper nouns

4.2.1 Personal names

Personal names in Classical Chinese have the surname-given name order. Yet the meanings of both *surname* and *given name* are not always clear. Before the Warring States period, the name of an aristocrat had two surnames: 姓 *xìng* 'ancestral clan name', 氏 *shì* 'branch lineage name'.

The merger between the two was eventually finished during the Han dynasty, possibly because of the collapse of the *fengjian* system: people who were not members of the Zhou dynasty nobility became politically active, and they likely had neither ancestral clan names nor branch lineage names. It was common for them to adopt their birthplaces as their surnames, which did not fit into the double surname system, leading to the elimination of the distinction between *xìng* and *shì*. Sima Qian himself does not clearly distinguish between the two.

- (1) 及生，名為政，姓趙氏

Besides the surnames, people also have 名 *míng* 'given name', the name given at birth. Still it is considered impolite to use *this* given name when having conversations with strangers or not-so-close friends: in the latter cases, another name, the 字 *zì* 'courtesy name (lit. character)' is used. A courtesy name, according to rites of Zhou, was to be given at the adult ceremony of a man, after which addressing him by his "regular" given name (i.e. his *míng*) was disrespectful.

- (2) 幼名，冠字

A courtesy name may be formed by prefixation: it was common before the Qin dynasty to first have a monosyllabic proper of the courtesy name and then prefix 子 (a respectful title) or a numeric index (伯 'first', 仲 'second', 叔 'third', 季 'last') expressing the man's birth order in his family.

For example, an important disciple of Confucius, usually known as 子路 in the modern literature, has 由 as his name given at birth (i.e. his *míng*): 子路 is his courtesy name, where 子 is a respectful title for a man. In *Analects*, Confucius usually calls him 由 because of their close relationship, while others, including the editors of *Analects*, call him 子路 (i.e. his *zì*). His *shì* is 仲, and therefore when both the surname and the given name are called, 子路 is known as 仲由.

(3) 子路聞之喜。子曰：「由也好勇過我，無所取材。」

(4) 仲由可使從政也與

The practice to have courtesy names rapidly ceased in the twentieth century, when, quite similar to how the distinction between the ancestral clan name and the branch lineage name disappeared, due to rapid modernization, scholar-officials lost their positions in government, and rural land owners lost their properties both in mainland China and in Taiwan due to violent or non-violent land reforms. The “lay” citizens and peasants, now at the center stage of economy or even politics, had never had serious exposure to the tradition of courtesy names, and it is predictable that no one cared about courtesy names anymore.

Moreover, besides the courtesy name, certain people may have art names, or 號 *hào* ‘lit. mark’. A person may have multiple art names. The great poet 李白, whose courtesy name is 太白 ‘lit. extreme-white’, had 青蓮居士 as his art name or *hào*.

4.2.2 Place names

TODO: e.g. 姑蘇

Chapter 5

Pronouns

Box 5.1: Third person pronouns

之 seems to be the accusative pronoun in Old Chinese. 其 seems to be the genitive pronoun, and may be a phonological fusion of 之 and a possessive marker.

我，吾，
在下陛下閣下

Chapter 6

Prepositions

Prepositional complements are also placed after the verb (1).

- (1) 君子博学于文

The peripheral arguments can be post-verbal (2, 3, 4) or pre-verbal (5, 6, 7), with the pre-verbal order gaining popularity as time went by. The linear order of peripheral arguments is similar to that in Mandarin (He 2005, pp. 286-287).

- (2) 侍饮于长者
(3) 孟孙问孝于我
(4) 祷尔于上下神祇
(5) 韩生南向坐
(6) 於人之罪无所忘
(7) 为人谋而不忠乎

Chapter 7

Verbal morphology

7.1 Stem derivation

7.1.1 “Nouns used as verbs”

The conventional term in Mandarin Chinese 名词作动词 ‘nouns used as verbs’ covers two phenomena, corresponding to multiple functions and zero derivation (Dixon 2010, § 11.3), and also the rare case of ad hoc re-categorization of a root.

7.1.1.1 Multiple functions

Some roots have both nominal and verbal uses, and there is usually some semantic connection between the interpretations of the two uses, but this is not regularly inferable. Here we consider some examples in Yang, Kong, and Zhou (1991):

- 楚 may mean ‘the Chu state’ or ‘do what Chu people do’.
- 床 may mean ‘bed’ or ‘settle down your bed or sleep on a bed’.
- 城 may mean ‘city, castle’ or ‘build a city’.

The interpretation of the verbal usage is usually *not* decided from the meaning of the root and that the root is used in a verbal environment; rather, it is instructed by the lexicon. Therefore, the verbal usage of 城市 only means ‘build a city’ although the ‘do city-related things’ reading in principle could make sense.

Therefore, roots like 城, 楚 and 床 have double functions: nominal and verbal, but the two functions are likely not related to each other by regular grammatical rules. This corresponds to the “multiple function” case in Dixon (2010, § 11.3). Moreover, what is stored in the lexicon is not the bare, non-categorized root 城, but one noun lexeme 城 ‘city’ that specifies its nominal usage and one verb lexeme ‘build a city’ that specifies its verbal usage, and other seemingly possible ways to categorize the root, although attested elsewhere, are ruled out by their absence in the lexicon.

The boundary between roots with double functions and roots undergoing zero derivation (see below) is somehow blurry, as the nominal and verbal uses of 城 and 床 still seem to show a common pattern and may be understood as a rare derivation. This blurriness leads many grammatical works on Classical Chinese to simply refer to the two phenomena uniformly as “nouns used as verbs”.

7.1.1.2 Zero derivation

In other cases the meaning of the verbal use of a root usually appearing in a nominal context is regularly derived from the nominal meaning. This is because although tropative or causative derivations in Classical Chinese are mainly verb-to-verb, they can also be applied to nouns. In this way, from 臣 ‘servant, official, minister’ we have the causative verbal usage ‘make sb. dependent on’, and from 客 ‘guest’ we have the tropative usage ‘consider sb. as a guest’. These verbal usages are nothing different from noun-to-verb derivation observed in other languages, so we regard the relevant phenomena as zero derivation as in Dixon (2010, § 11.3).

In zero derivation, the meaning of the nominal usage has to be recorded in the lexicon, the meaning of the verbal usage can be automatically decided from the derivation rule. These derivations are however not completely regular and not for every word: the lexicon also controls whether a derivational rule applies.

7.1.1.3 Ad hoc re-categorization

There are sporadic verbal usages of nouns that are almost never attested elsewhere, like 军 in 沛公军霸上. This means that ad hoc re-categorization of roots is possible in Classical Chinese, and the meaning is to be decided from the context. This is also possible in informal English¹ but usually not accepted in formal texts. Alleged ad hoc categorized Classical Chinese roots are indeed a possibility, after all, although their frequency is not high enough and cannot be exaggerated to be the norm rather than the exception.

- (1) 大楚兴，陈胜王
- (2) 然而不王者，未之有也

7.1.2 Numeral-to-verb derivation

- (3) 六王毕，四海一

7.1.3 Adjective-to-verb derivation

Box 7.1: Adjective-to-verb derivation

则其好游者不能穷也
山多石，少土
亲贤臣，远小人

7.2 Complex predication

Box 7.2: Complex predication

What matters: synchronic, or diachronic? Relation to valency alternation (no relation when diachronic)

¹As in *I might [guinea pig] it for you*. A longer example is a historic event being “*Sappho and her friend*”d.

Chapter 8

Verb frames

8.1 Intransitive constructions

8.1.1 Action verbs

8.1.2 Stative and change-of-state verbs

Known in Mei (2015) as 内动

A BE verb describes a state; a BECOME verb describes the change of a state. In both types of verb frames, the sole argument is a *theme*: the situation happening to it just happens, and usually it does not have much control over it nor any volition to trigger it (Li 2004, p. 345; Mei 2015, p. 275).

In Classical Chinese, just like in other languages, BECOME/BE verbs often have established causative usages, forming CAUSE-BECOME/BE argument structures with the *causer* argument being the subject and the *theme* argument being internal (§ 9.1.1). When a causer is absent, the structure of the clause is comparable to what sometimes is known as the middle voice in English (e.g. *the door opened*; cf. the transitive CAUSE-BECOME *I opened the door*). (1) is an instance: in its CAUSE-BE usage (2), the argument that is described as weak is an internal argument that appears after the verb, but in (1), the argument that is described as weak is the *subject*: the internal theme argument gets promoted to the subject position.

- (1) 秦强而赵弱

The “middle voice” construction exemplified in (1) (known as 内动 in Mei (2015)) has a subject, which corresponds to the argument that is the object in the CAUSE-BECOME/BE construction (i.e. the internal argument).

8.1.2.1 The alternation between BE and BECOME

Alternation between BE and BECOME verb frames is natural. Some BECOME verbs however do not have BE counterparts.

8.1.2.2 Impersonal stative verb

It is possible for the argument of a stative verb to be *not* promoted to the subject position. (2, 3).

- (2) 蟋蟀 鸣
? chirp
‘??? chirp.’ (礼记·月令)

- (3) [五 月]_{temporal} [鸣]_{predicate} [蛎]_{internal argument}
 five month chirp cicada
 ‘In the fifth (lunar) month, cicadas chirp.’

(3) can be analyzed as a situation if we assume that 鸣 is a BE verb, denoting a state where bugs continue to make noise, and that the sole argument 蛎 remains in-situ and is not promoted to the subject position.

No other analysis is available: for instance a DO verb can never have such a behavior (Mei 2015, p. 351).

8.2 Monotransitive constructions

8.2.1 Action verbs

The subject of a DO clause usually has to be animate, because it voluntarily initiates the event described by the clause (4). The subject is an *agent*, as opposed to a *causer* (§ 9.1) or a *theme* (§ 8.1.2).

Mei (2015, p. 272) lists some criteria to distinguish a transitive DO verb from a transitive CAUSE verb (§ 9.1.1).

- (4) 桓公杀公子纠

8.2.1.1 S=A labile verbs

- (5) 昭王反国
 (6) 右师反

8.2.2 Causative verbs

Mei (2015, p. 272) summarizes the main difference between transitive, DO action verbs and causative verbs.

8.2.2.1 S=P labile verbs

- (7) 大夫溃莒
 (8) 莒溃

8.2.2.2 Lexicalized synthetic causative construction

Some CAUSE verbs are fossilized, and do not have clear intransitive counterparts i.e. the un-causativized counterparts. For instance, 伤 ‘hurt’ typically is a state transition verb meaning body, etc. being hurt, and it also has a causative (i.e. CAUSE-BECOME) meaning (‘make ... hurt’). The CAUSE-BECOME verb frame of 伤 however has gained a separate lexicalized meaning that can’t be transparently inferred from the meaning of the BECOME usage: it can mean ‘let ... be damaged’, in which the object is not necessarily body or a person. This usage of 伤 has no BECOME or other intransitive counterpart. The absence of a BECOME counterpart can be proven by the ability for this fossilized figurative usage of 伤 to undergo “passivization” (9), which is otherwise not possible (§ 9.2.1).

- (9) 女红伤则寒之原也

Verbs like 伤 in like (9) can easily be reanalyzed as DO verbs. This is likely a diachronic path of the creation of DO verbs. The verb 败 for example seems to be originally a BECOME verb (‘to get corrupted’) and later gained a specific meaning of ‘to defeat’ in its CAUSE-BECOME usage, which eventually evolved into a DO usage (Mei 2015, p. 285).

8.2.2.3 Causative verbs with non-agentive subjects

Some verbs license subjects that look like arguments of prototypical BECOME or BE verbs: the subject may be animate but it does not volitionally trigger the event. The situation “just happens to be the case”, and the subject can be described as a *theme* and not an *agent*. What sets them apart from prototypical BECOME or BE verbs in § 8.1.2 is the fact that the subject seems quite unlike an internal argument. In (10), the subject 火 ‘fire’ is definitely a theme and not an agent: the fire does not get to *decide* if it burns the flag (Mei 2015, p. 276). Still the clause is not a prototypical BECOME one as there is an internal argument 其旗 in it, and the theme 火 is an *external* theme (Mei 2015, p. 353). These verbs therefore have difficulties participating in synthetic causativization (§ 9.1.1).

- (10) 火焚其旗

8.3 Ditransitive constructions

8.3.1 Verbs governing a locative object

Compare the prepositional (11) and the double-object example (12). The omission seems to be in line with the omission of the preposition in post-verbal locative arguments (cf. 13).

- (11) 有献不死之药于荆王者

- (12) 请献盆缶秦王

- (13) [大-王]_{subject: NP} 见 [臣]_{object: NP} [列 观]_{locative: NP}
 great-king see servant/minister regular palace
 ‘Your Majesty see (your) servant (i.e. me) at a regular palace.’

8.3.2 Experience verbs

Some experience verbs, mostly verbs about emotions, behave like BECOME verbs (Mei 2015, p. 273): when used as transitive verbs, the subject does not look quite agentive and the clause is likely causative (15), and when used as intransitive verbs, there is a clear internal change-of-state meaning (14).

- (14) 孔子成春秋，而乱臣贼子惧 (孟子·滕文公章句下)

- (15) 惧之以怒 (左传·昭公十三年)

On the other hand, perception verbs (e.g. 见 ‘look’) and cognition verbs (e.g. 知 ‘know’) are often transitive, and therefore are not compatible with the synthetic causative construction (§ 9.1.1; Mei 2015, p. 274). Intuitively, these verbs are DO-like according to the criteria listed in § 8.2.1. For instance, they can appear in 所 construction (16).

- (16) 異乎吾所聞

Certain perception verbs however have developed a figurative, fossilized meaning, and when intransitivized, can participate in synthetic causativization (§ 9.1.1, 3). This possibility indicates that these fossilized usages are BECOME- or BE-like: 见 ‘meet formally’ therefore means ‘in the state of regularly meeting an important figure’.

8.4 Experiential valency increasing

8.4.1 The affective constructions

Classical Chinese has two affective constructions, in which the subject is an experiencer suffering something bad from the situation described by the latter (Mei 2015, pp. 354-358).

The first affective construction simply attaches an experiencer to an argument structure (17). In (17a), 亡 appears as a BECOME verb (§ 8.1.2): it is intransitive and its subject, the *Odes*, did not have control over its being ignored. In (17b), a new argument – the experiencer subject – is introduced to the argument structure of 亡: the meaning of the sentence is ‘the shepherds suffered from the sheeps getting lost.’

- (17) a. 亡 as a BECOME verb
[诗]_{subject,theme: NP} [亡]_{BECOME} 然后 春秋 作
poem **get.lost** then Spring-Autumn compose
‘The *Odes* got lost, and then the *Spring and Autumn Annals* was composed.’
- b. 亡 as a AFFECTIVE-BECOME verb
[二 人]_{subject: NP} [相 与 牧 羊, 而 俱 亡]_{AFFECTIVE-BECOME}
two person mutually go.together herd sheep CONJ all **get.lost**
其 羊]_{coordinated VP}
poss sheep
‘Two people herded their sheep together, and they both lost their sheep (lit. suffer from their sheep’s missing).’

The second affective construction *obligatorily* has an object, which is in possession of the subject. The meaning of (18), for example, is that Confucius was frustrated by the fact that his tree was cut in Song.

The fact that the object (树 and 肋 here) should not contain any possessive markers and has to be interpreted as something being possessed by the subject suggests that the second affective construction is an external possession construction.

(18) 吾再逐於鲁，伐树於宋

(19) 范雎折肋於魏

Box 8.1: External possession

External possession as subject

8.5 Applicative constructions

Box 8.2: Applicative constructions

benefactive; the claim that double object constructions are similar to benefactive constructions; what object gets passivized. See Mei (2015, p. 421).

(20) 子使漆雕開仕

(21) 雍也可使南面

8.6 Direct quotation

《詩》云：『如切如磋，如琢如磨。』

8.7 Control

In (22), the constituent introduced by 曰 is a proper name, and clauses with 曰 as the main verb where the subject is a person and the object is a proper name are rare if not impossible. If 曰 in (22) is understood as a marker of a direct quotation, however, the sentence makes sense: the second argument has a *metalinguistic* usage, whose semantic interpretation is the quoted 灵台 itself, without referring to anything in the real world.

(22) [谓]_{verb} [其台]_{object: NP} [曰 [灵台]_{proper name}]??

Therefore, in Classical Chinese, 曰 is both used as a lexical verb (18) and a grammaticalized marker of direct quotations. What is quoted can be a sentence (19) or a noun phrase (22). Under this analysis, examples like (23) are probably clauses with nominal predicates, with the predicate being a direct quotation.

(23) 其名曰觚

Box 8.3: In search of auxiliaries?

See e.g. [here](#).

- 克 “能” “堪” “可” “可以” “可得” “得” “足” “足以” 等
- 欲 “肯” “將” “宁” “敢” “忍” “愿” “屑” “慙 (yìn)” 等，其中 “敢” “忍” “屑” 通常用于否定句，“肯” “(不) 敢” “(不) 忍” “(不) 屑” 沿用至今。“欲” “將” “愿” “慙” 表示主观的希望或打算，可以翻译成 “希望” 或 “打算”。例如：
- 当 “如” “宜” “任” “合” “应”，其中 “合” “应”
-

能 does not seem to be a prototypical auxiliary. The main argument against its status as an authentic auxiliary is the prevalence of sequences like 不能不 (as in 人主不能不有遊觀安燕之時). Here the two negations mean ‘not doing ...is impossible’, and the structure is routinely used to emphasize the core verb phrase (VP). The structure of these examples seems to be better analyzed as a complement clause construction, as nowhere else in Classical Chinese can we find two identical negators in a single clause. (非不 is possible, but not 不不)

cf. Negation in Cartography

Chapter 9

Valency alternation

9.1 Causative constructions

A *causer* makes a situation to be the case, but does not always do so intentionally. It can therefore be inanimate, as opposed to how an *agent* behaves (§ 8.2.1).

We can divide causative constructions in Classical Chinese into synthetic and analytic ones. In the synthetic causative construction, there is only one verb in the surface form: the causative valency alternation is supposedly marked by a prefix *s-, which is invisible in the written texts but is reflected by tonal changes of the verb. If a root develops a lexicalized usage in the synthetic causative construction, then CAUSE verb is formed.

9.1.1 Synthetic causative

The synthetic causative construction applies to existing argument structures, or sometimes bare roots. The synthetic causative construction cannot be applied to a DO construction: the reason is probably because a DO construction is too “big”, already having a full-fledged wannabe subject (Mei 2015, p. 363-364).

On the other hand, the syntactic causative construction can be applied to a pseudo-passive structure (1) and BECOME or BE (2) argument structures. Certain intransitivized experience verbs, possibly having an argument structure comparable to a BECOME/DO verb (§ 8.3.2) with a wannabe subject also have causative usages (3), but their transitive counterparts are never compatible with the synthetic causative construction (Mei 2015, p. 274).

Finally, the synthetic causative construction can be directly applied to a root (4): the word 妻 ‘wife’ is sometimes used as a verb, meaning ‘to marry daughter to ...’, inconsistent with the meaning of (4). Therefore, in (4), 妻 is ad hoc categorized into a CAUSE verb, its usual verbal usage being irrelevant here.

- (1) 是 夭 子蛮, 杀 御叔……
this die.young NAME kill NAME
‘This woman made Ziman die at a young age, and got Yushu killed...’
- (2) 以 弱 天下 之 民
PURPOSE weak world GEN people
‘...to weaken the people.’
- (3) 子尾 见 疆
NAME formally.visit NAME
‘Ziwei let Jiang formally visit (with Xuanzi).’ (左传·昭公二年)
- (4) 妻帝之二女

The labile S/O alternation between the BE/BECOME usage and the CAUSE-BE/BECOME usage is quite regular in Classical Chinese; verbs allowing this alternation are sometimes known as *ergative verbs* (Mei 2015, p. 378), although the phenomenon is about the core argument structure and has nothing to do with ergativity in alignment. It should be noted that not all BECOME/BE verbs are compatible with the synthetic causative construction. For instance, 鳴 ‘chirp’ in (3) does not have a transitive CAUSE-BE usage. More examples are given in Mei (2015, p. 276). On the other hand, some clauses that look like CAUSE-BE/BECOME clauses actually do not have BE or BECOME counterparts (§ 8.2.2.2).

9.1.2 Analytic causative

Classical Chinese has an analytic construction to express the causative meaning. In (5), the word 使 is applied to the stative structure 渚者居中原 ‘people living on small lands reside inland’, meaning ‘let people living on small lands in water live inland’.

- (5) 不 使 [渚者]_{shared object: NP} [居]_{BE} [中-原]_{locative object: NP}
 NEG let small.land.in.water NMLZ reside middle-land
 ‘...do not let people living on small lands in water live inland’

The word 使 can be replaced by 令 (6) or 俾 (7) (Mei 2015, p. 376). Note that in (6), 令 seems to be applied to a DO argument structure.

- (6) 令军勿敢犯
 (7) 俾民不迷

A question is whether 使 is a lexical verb, or just a non-incorporated causative marker. The fact that (6) involves a DO verb deviates from the behavior of the synthetic causative construction, which is not compatible with DO verbs (§ 9.1.1). Further, the fact that we can choose among 使, 令 and 俾 is rather unusual for a grammatical marker: we also note that 使 retains the meaning of ‘send sb. to do sth.’ and 令 retains the meaning of ‘command sb. to do sth.’ Therefore, we consider 使, 令 and 俾 to be *lexical verbs*, and not grammaticalized causative markers. What is grammaticalized is the causative clause with the form of V₁ NP VP, with V₁ being one of these verbs.

9.2 “Passive” constructions

9.2.1 The agent-less pseudo-passive construction

What is often known as the passive in Classical Chinese is not really a passive construction comparable to the English or Latin passive. The main problem is the lack of a grammaticalized way to say the agent: in a “true” passive construction, the original subject is somehow demoted (represented by the appearance of *by* in English or the ablative case in Latin) and sometimes omitted, and an internal argument is promoted to the subject position, while the so-called “passive” constructions in Classical Chinese – which we name the *pseudo-passive* here – involve an agent that is *obligatorily removed* (Mei 2015, p. 287-289).

9.2.1.1 Passivization vs. information packaging

In a language that allows multiple topicalization and focalization, whether omission of the agent is to be analyzed as argument structure alternation instead of information packaging needs discussion. One may want to analyze the promotion of the patient argument to the

subject position as pro-drop of the agent plus topicalization of the patient. This analysis essentially eliminates the necessity of the concept of subject in this construction. However, we contend that this analysis is not viable.

First, we note that Classical Chinese has explicitly marked passive constructions like the 見-construction. Their existence cannot be easily captured within an information structure-only framework.

Second, we have both passivization after causativization and causativization after passivization (§ 9.2.1.2), and the semantics and the word order in the two constructions are regularly determined. This strongly indicates that we are encountering valency alternation.

Third, information structure-driven omission of arguments, i.e. pro-drop in Classical Chinese mainly applies to arguments representing *known* entities. Yet the agent omitted in a pseudo-passive construction is frequently *unknown*.

9.2.1.2 Relation with causative constructions

It is not frequent but not impossible to apply pseudo-passivization to a causative construction (Mei 2015, p. 283). Suppose we have a bivalence causative construction. If pseudo-passivization is applied to it, we are to suppress the external argument and let the internal argument to be the subject. But such a bivalence causative construction usually has a CAUSE-BECOME structure, and removing the causer leaves us a clause that looks just like a BECOME clause. So there are two competing analyses, and since usually if a verb root is lexically licensed to head a CAUSE-BECOME clause, then its usage in a BECOME clause is also in the lexicon, the simpler BECOME analysis is preferred. However, where this preference is eliminated, pseudo-passivization of a causative construction is possible (Mei 2015, pp. 284,370-372; § 8.2.2.2).

Given the strict conditions of pseudo-passivization *after* causativization, alternate application of pseudo-passivization and causativization that results in a PASS-CAUSE-PASS-... argument structure is not attested, although such a combination is not semantically unfeasible (e.g. consider ‘someone is made to be made to do sth.’), and is occasionally attested, probably in a playful way, in living languages like modern Japanese (8). This indicates that the argument structure in Classical Chinese is subject to considerable lexicalization and valency alternation constructions are not completely productive. In this particular case of the impossibility of a CAUS-PASS-CAUS chain, what is revealed probably also includes the decay of productive voice affixes in Old Chinese.

- (8) 見つかる と 削除させられさせられる

mitsukar-u to sakujo-sase-rare-sase-rare-ru
see-PLAIN.NONPAST COND delete-CAUS-PASS-CAUS-PASS-PLAIN.NONPAST

‘If it is discovered, someone will be made to be made to delete it.’ (Modern Japanese, from Internet.)

9.2.1.3 The case with multiple internal arguments

When the pseudo-passive construction is applied to double object clauses, it is the *recipient* that is promoted to the subject position (9) (Mei 2015, p. 421). On the other hand, when the pseudo-passive construction is applied to the corresponding prepositional argument construction, it is the *theme* that is promoted to the subject position (10). These phenomena establish a hierarchy of *pivotality* of arguments (§ 2.2.4.2).

- (9) 诸侯，赐弓矢然后征

- (10) 药言先献于贵，然后闻于卑

TODO: 杀御叔

9.2.2 Tropative

Tropative is a construction which attaches an experiencer to a BE argument structure, with the meaning being ‘*A* consider *B* to be ...’ (Mei 2015, pp. 413-414) The Classical Chinese tropative is actually not limited to stative verbs: it also applies to nouns. It is however not likely that this construction comes from transformation of the nominal predicate construction (§ 2.2.1). The main difference is that in the nominal predicate construction, the predicate is a noun *phrase*, but the tropative construction never takes a nominal predicate as input.

Chapter 10

Nominal predication

Box 10.1: The possessive nominal predicate construction

It seems the predicate in the possessive nominal predicate construction can never be a bare noun without any modification. The modification can be a numeral or an attributive.

- (1) 王六军，大国三军
- (2) 秦王[为人]_{Box 18.1}，蜂准，长目，挚鸟膺，豺声，少恩而虎狼心

Another problem is that the 者-也 construction seems to be incompatible with the possessive nominal predicate.

Chapter 11

Tense, aspect, modality

11.1 Time adverbs

It is probably more appropriate to recognize these TAM markers as (semi-)lexical adverbs instead of auxiliaries or particles, because the class of TAM markers seems to be open during the Classical period. We believe that they have been incorporated into a grammaticalized TAM system, because depending on their meanings, they have a fixed linear order, consistently with the order of TAM markers observed cross-linguistically (§ 2.2.3). For instance, from (1), we observe that the epistemic adverb 或 appears before the future adverb 將. Still, multiple time adverbs are rare in truly Classical texts, and whether an adverb is a part of the TAM system or a temporal peripheral argument like *at that time* sometimes has to be determined by its semantics.

Box 11.1: List of time adverbs

list

- Are some of them peripheral arguments?

11.1.1 Subject-oriented adverbs

The pre-verbal particle 或 seems to be an epistemic modality marker. In (1), we see the coexistence of the epistemic 或 and a future adverb 將.

- (1) 或將豐之，不亦難乎

必 indicates the speaker's strong confidence towards the statement being made; similar adverbs include 固, 實, and 誠 (Li 2004, pp. 376-377). These adverbs also appear before the future 將, further confirming their status as epistemic modality or evidentiality markers.

- (2) 知惠之必將至也
(3) 諾，吾固將圖之

We can further observe both the linear order 固 > 必 and the reverse.

- (4) 固必通乎性命之情者
(5) 將欲弱之，必固強之

11.1.2 The tense adverbs

嘗 seems to be a marker of the simple past.

The concept of future events can be expressed by 將, but it is not compatible with 嘗. On the other hand, in English, we have *would do sth.* or even *would have done sth.* The incompatibility of 嘗 and 將 is evidence for the syntactic distinction between how future is expressed in English and in Classical Chinese: in English, we have both a past/non-past distinction and a future/non-future distinction, beyond the standard Reichenbach system (Vikner 1985), while in Classical Chinese, the past future tense is non-existent and we have a tripartite past/present (related to 矣; § 11.3.1)/future distinction.

11.1.3 The modality adverbs

(6) 又將能忍吾子乎

11.2 (Semi-)auxiliaries

In English, it is possible to have both *want to be able to do sth.* and *be able to want to do sth.*¹ A corpus search reveals that 可 ‘should, ought to’ always appears before clearly lexical verbs, while the reverse order is not attested.

(7) 見其可欲也，則必前後慮其可惡也者

Further, consider 罪當可赦: 當 > 可.

11.3 The role of sentence final particles

11.3.1 矣 and the tense system

The appearance of the sentence final particle 矣 seems to have a tense effect. When 矣 appears and there are no other TAM markers, the relation between the event time and the speech time is still largely arbitrary, but the event always seems to have some relevance to the speech time: the meaning of English *somebody did something* or *somebody had done something* is not admissible. It seems that these observations can be summarized as follows: the grammar of Classical Chinese does (although implicitly) recognize Reichenbach’s speech time, reference time and action time, and when 矣 appears, the latter two are identical, hence prohibiting the simple past meaning (Mei 2015, pp. 437-446). However, after considering its combinations with other markers, we do not consider 矣 itself to be a tense marker (§ 13.2).

¹This may sound awkward, but it makes sense in certain contexts, like *you have to be a special kind of nerd to be able to want to drive 28 hours for a video game.*

Chapter 12

Negation

Chapter 13

Sentence final particles

13.1 Sentence final particles as discourse markers

Classical sentence final particles have a variety of functions. They may mark the interrogative force (1), a judgmental meaning (2), and aspectual values (3).

- (1) 大车无輶，小车无輶，其何以行之哉
- (2) 人而无信，不知其可也
- (3) 温故而知新，可以为师矣

It seems a sentence final particle can be shared by two conjuncts.

- (4) 虎者，戾虫；人者，甘饵也

In early texts, a sentence final particle can be inserted after the main verb.

- (5) 巧言令色，鲜矣仁

At first glance, the sentence final particle 矣 marks the perfect aspect (6), while 也 is for clauses describing something happening regularly, not a single concrete event (7) (Mei 2015, pp. 443-445).

- (6) 余助苗长矣
- (7) 将发命也

矣 and 也, however, are different from prototypical aspect markers in several aspects. First, 矣 appears predominantly in direct quotations in Classical texts, which suggests that it has conversational functions. 也 frequently appears in narratives as a part of the judgmental construction (§ 2.2.5.1), which lacks TAM marking, and 也 cannot be a prototypical aspect marker in that context. Second, it seems that 矣 and 也 can be shared by two coordinated conjuncts with different subjects (§ ??), which is rather unusual for an aspect marker.

Therefore, the grammatical category corresponding to 矣 and 也, whatever it is, is not a typical TAM category, and hence we disagree with Mei (2015)'s analysis. Its scope is wider than TAM categories: a TAM category is in relation with a nucleus clause, while 矣 and 也 are in relation with a sentence, i.e. an arbitrarily complex clause that is one utterance in a conversational context. This is consistent with the usual analysis of sentence final particles in modern Standard Mandarin (Paul 2014b; Pan 2021).

- (8) 亦各言其志也已矣

Box 13.1: Alternative analysis

Zhu (2009, p. 233) acknowledges the widespread analysis that sentence final particles are in relation with the whole sentence, not the nucleus clause, but insists that certain sentence final particles are a part of the predicate. Yet no convincing argument is provided. Among the three distributional classes he recognizes in Mandarin, two (marking the interrogative/imperative force, and attitude of the speaker) are uncontroversially attached to *sentences* and not nucleus clauses. The remaining class, which is called the “tense” class by Zhu (2009) and is structurally the innermost, resembles the class of 矣 and 也 discussed here, seems to be forbidden in most embedded clauses (Deng 2010), just like the other two classes do. Therefore all the three classes of Mandarin sentence final particles described in Zhu (2009) are indeed in relation with the sentence and not the nucleus clause, which is consistent with the structural status of sentence final particles in Classical Chinese.

We also note that it is not completely impossible for two independent nucleus clauses to share one TAM marker. In Japhug, for example, a series of nucleus clauses with different subjects can be coordinated with the TAM categories being marked at the end of the compound clause (Jacques 2021, pp. 1090-1091). However, Japhug lacks clear clause-level subject (Jacques 2021, § 2.5.3), and therefore coordinated nucleus clauses with a shared TAM marker but different “subjects” is less strange in Japhug than it is in Classical Chinese: in the latter, we have a well-defined clausal pivot grammatical relation (§ 2.2.4.3) whose scope is over all TAM categories, making clauses sharing the TAM marker but not subjects rather unusual, but in the former this is probably not the case.

Mei (2015, pp. 443-445) relies solely on semantic criteria. Although we do not believe his analysis of 也 and 矣 as tense markers is completely correct, the two clearly have non-trivial interaction with TAM categories, a phenomenon also observed in Mandarin, where the lowest sentence final particle has access to TAM categories of the nucleus clause (Paul 2014a, p. 258).

13.2 矣

矣 is not a tense marker Besides the fact that sentence final particles are discourse-oriented and therefore not prototypical TAM markers, we also note the following two facts.

First, 矣 is compatible with both 將 and 嘗 (9, 10; § 11.1.2), which probably means it is not directly related to the tense system.

(9) 難將至矣

(10) 牛山之木嘗美矣

Second, 矣 seems to have a weak exclamative function. For example, in (11), the speaker has something to say but has waited for quite a while for a change to say it. In (12), Confucius is impressed by how well cultivated 韶乐 ‘Shao music’ is.

(11) 吾欲言之久矣

(12) 尽美矣

We therefore conclude that 矣 is not a tense marker, and instead analyze it as a “sentential aspect” marker, which informs the listener that a new piece of information arrives. This is consistent with the function of the sentence final 了 in modern Standard Chinese.

Box 13.2: Alternative analysis

In Ch. 11 of Mei (2015), it is claimed that 矣 is a prototypical tense marker. We agree with Mei (2015) in that Classical Chinese has a (rather weak) tense system, and we basically reiterate his arguments above. His arguments for the claim that 矣 itself is a tense marker however we do not consider valid. On p. 447, he argues that 矣 appears in imperative sentences and therefore cannot be related to aspectual categories. But the category of tense also rarely appears in imperative sentences. On the other hand, the sentential aspect category is essentially a discourse-level category, and has no inherent conflict with the imperative speech act. On p. 448, he notes that in coordination, when 矣 appears at the end of the first clause, it sets up a reference time for the following clauses. Yet he states clearly that the reference time is actually the event time of the first clause, whose relation with the speech time is *not* specified by 矣.

On the other hand, if we accept the analysis that 矣 hints at new information that the speaker wants to express to the listener at the speech time, then we can easily understand why 矣, because a generic fact like *the Earth goes around the Sun* rarely is new information, and a piece of new information usually involves a specific situation at a specific time, which is inevitably compared with the speech time, and hence the tense system. The fact that 矣 is often related to the present tense or the present perfect tense may simply be because of the absence of other TAM markers: cf. (9, 10).

Other sentence final particles, like 也, do not have this effect: hence an alternation between narrations about events and descriptions on what these events mean on the battlefield (Mei 2015, pp. 444-445).

Box 13.3: The relation between topic and sentential aspect

A question is whether we can claim that the category of the sentential aspect is somehow higher than the category of topic. If so, then the sentential aspect belongs to CP. See Mandarin peripheral construals at the syntax-discourse interface, page 30. 是北京昨天下雪了: the focus can be on 北京 and 昨天下雪了 modifies 北京, or 了 modifies

Mei (2015, p. 445) and Pulleyblank (1995, p. 19) both claim that form 也矣 is impossible. It turns out that it is rare, but not non-existent. It however has to be admitted that 也已矣 is much more frequent than 也矣. Pulleyblank (1995, p. 20) analyzes 已 (when used as a sentence final particle) as a fusion of 也 and 矣.

(13) 是善惡之分也矣

Box 13.4: Other combinations

寡人之於國也，盡心焉耳矣

Chapter 14

Clause types

Box 14.1: Clause types

In Mei (2015, p. 131), he classifies clauses into 说明句, 描写句, and 叙事句. The classification is comparable to that given in <http://area.hcjh.tn.edu.tw/noise/hcjh-ca/4-b.htm#0303>. Mei doesn't mention on which basis he makes this distinction. In the latter source, it seems the distinction is made based on the type of the predicate. Thus a 描写句 is a stative (adjectival?) clause, and a 判断句 is a nominal predicate construction, and a 叙事句 is a verbal predicate construction that is not a 描写句. So what does Mei mean by 说明句? The term appears in Li (2004) as well.

We can go to places where he mentions the term. p. 445: 矣 is for 叙事, and 也 is for 说明. pp. 264-265: 事件句 (叙事句) 和非事件句 (描写句和说明句) The distinction is also mentioned in <http://paper.wenweipo.com/2018/02/14/ED1802140024.htm>

So it's related to the event structure. We need to know where the event structure resides in the vP-TP-CP hierarchy. Particularly, we need to identify *where* the category of this distinction lies. I think probably that's related to the aspect: consider the distinction between a habitual clause and a prototypical "event" clause.

The distinction has syntactic significance. We note that certain topicalization constructions seem to be only compatible with 说明句 (Box 18.1).

Box 14.2: Adjectival predication of a non-finite clause?

Consider 其為人也，發憤忘食，樂以忘憂，不知老之將至云爾: the subject of 發憤忘食，樂以忘憂，不知老之將至云爾 is clearly 其 i.e. Confucius. But then consider the fact that the subject *cannot* appear after 也. This seems to indicate that we are dealing with some sort of subject sharing construction.

Also, consider 為文斐然可觀矣: here 斐然客觀 semantically modifies 文?

See also Box 18.1.

Chapter 15

Information structure

(1) 单父人吕公……吕公者，好相人，……

Chapter 16

Subordination

The marker 乃 is also available as a marker of the consequence clause (Mei 2015, p. 87).

Other markers for the condition clause are also available (Mei 2015, Ch. 3). We note that the marker 若 is able to appear *after* the subject of the condition clause (Mei 2015, p. 94).

condition (1) or temporal clause (2) can be a gerundive one (§ 2.2.6.2).

(1) 我之不德，民將弃我

(2) 臣之壮也，犹不如人

In some condition clauses, the marker 而, instead of the otherwise genitive marker 之, appears between the subject and the predicate, forming a clause type that is not gerundive and only appears as an irrealis condition clause (Mei 2015, pp. 100-102).

Chapter 17

Genres and formulae

17.1 Idioms

- (1) 非请勿入

17.2 Classical-like formal documents

17.2.1 Background

Box 17.1: Background

- 各代公文
- 晚清民国

17.2.2 Pseudo-Classical documents

The text shown in (2) is an article from the Chinese version of the Geneva Conventions, an example of the pseudo-Classical Chinese used in official documents. The text is Classical-like when we inspect the grammatical markers appearing in it: 之 in place of 的 as a relative clause marker and also a possessive marker, 是 in place of 这 as a demonstrative, and 彼等 in place of 他们 as a third person pronoun. Yet several features can be immediately noticed that deviate from the Classical standard.

- (2) 战俘不得放弃本公约或上条所述之特别协定——如其订有是项协定——所赋予彼等权利之一部或全部

First, polysyllabic content words are prevalent. The verb 放弃 ‘to give up’ is not unheard of in Classical texts (cf. 3), but it is usually analyzable as a verb-level coordination of 放 ‘to put down’ and 弃 ‘to abandon’. 放弃 as a whole with the meaning of giving up something abstract is a feature of Modern Standard Mandarin, where 弃 is no longer acceptable when used alone as a verb, which is indeed the case in the Chinese version of the Geneva Conventions.

Nouns like 战俘 or 协定 are never attested in common Classical texts. Their structures can be explained with the Classical grammar. Still, the fact that roots within them do not seem to appear freely suggests that the translator likely had the lexicon of Mandarin in their mind.

- (3) 放棄詩書，極意聲色

(TODO: add the second pseudo-Classical feature.)

It is not impossible for Classical grammar to appear in a pseudo-Classical document, though.

- (4) 有组织之抵抗运动人员之在其本国领土内外活动者

17.2.3 Formulae

Box 17.2: Formulae

- 此令

17.3 Traditional letters

Box 17.3: Formulae

- 钧鉴
- 谨呈
- 母亲大人膝下，敬稟者，日前寄上海婴照片一张，想已收到

The last example can be parsed with Classical grammar: ‘At the knees of my mother, what is respectively reported below is: ...’ Yet the formula X鉴 is kind of hard to parse. Further, consider 道席 in 某公道席: it is similar to an epithet, or something like *your Majesty*.

So we have three types of 提称语: one is related to the request that the recipient reads the letter (the X鉴 formula), one is related to the profession or status of the recipient (like 道席), and the other is essentially a locative expression (膝下). The origin of these formulae remains unknown.

Chapter 18

Discussions on quirky examples

Box 18.1: A之于B也

- (1) 寡人之于国也，尽心焉耳矣

The original structure is probably 寡人于国尽心. The fact that 寡人 and 于国 are subtracted from the nucleus clause is rather unusual. And similar examples exist. Consider 史记夫乐之与音，相近而不同. The original structure is 乐与音，相近而不同, and the insertion of 之 is rather strange.

- (2) 其为人也，发愤忘食，乐以忘忧，不知老之将至云尔

A之于B也 (or A之为B也), predicate, or A为B也, predicate. The structure seems to be parallel to the English *I, as a concerned citizen, want to emphasize that ...*, where *as a concerned citizen* obligatorily modifies the subject *I*. 其为人也 here seems to be a *frame*, somehow comparable to the “global” temporal or locational phrase. Another issue is that the sentence seems to be unable to represent a specific event: *昨日，孔子为人也，发愤忘食, while we have *yesterday, as a concerned citizen, I...*

- (3) 水之积也不厚，则其负大舟也无力。

- (4) [良人者]_{subject} [所仰望而终身]_{predicate} 也

It seems subjects of ordinary verbal clauses cannot be topicalized (5), but if the verb phrase is emphasized, topicalization is possible.

- (5) a. 三王既以定法度
b. *三王，既以定法度

- (6) 秦，虎狼之国，不可信，不如毋行

What's the structure of the sentence below?

- (7) 將者，欲伐而未成，见其臣尚可以谏，而季氏尚可以止也

In the example below, the subject of the second clause is ‘other people’, and yet it seems the three clauses share a topic (i.e. 人).

- (8) 人而不仁，疾之已甚，亂也

What's the structure of the sentence below? ‘But as for Qiu, is it about state affairs?’

- (9) 唯求則非邦也與

The sentence below seems to be due to omission: 宗庙会同，非诸侯而何为之？

- (10) [宗庙会同]_i，非诸侯而何 —_{main verb} —_i
- (11) 赤也为之小，孰能为之大
- (12) 其为人也孝弟，而好犯上者
- (13) 且夫水之积也不厚，则其负大舟也无力
- (14) (人)如礼何
- (15) 如之奈何
- (16) 父母唯其疾之忧

References

- Aldridge, Edith. “Cliticization and Old Chinese word order”. In: *LSA Annual Meeting Extended Abstracts*. Vol. 1. 2010, pp. 7–1.
- Aldridge, Edith. “Generative approaches to ergativity”. In: *Language and Linguistics Compass* 2.5 (2008), pp. 966–995.
- Aldridge, Edith. “Old Chinese determiner zhe”. In: *Historical syntax and linguistic theory* 18 (2009), pp. 233–250.
- Boston, Marisa Ferrara, Hale, John T, and Kuhlmann, Marco. “Dependency structures derived from minimalist grammars”. In: *The mathematics of language*. Springer, 2009, pp. 1–12.
- Bruening, Benjamin. “The lexicalist hypothesis: Both wrong and superfluous”. In: *Language* 94.1 (2018), pp. 1–42.
- Cinque, Guglielmo. *Adverbs and functional heads: A cross-linguistic perspective*. Oxford University Press, 1999.
- Danckaert, Lieven. “On the left periphery of Latin embedded clauses”. PhD thesis. Ghent University, 2011.
- Danckaert, Lieven. *The development of Latin clause structure: A study of the extended verb phrase*. Vol. 24. Oxford University Press, 2017.
- Deal, Amy Rose. “External possession and possessor raising”. In: *The Wiley Blackwell companion to syntax* (2017), pp. 1–32.
- Deng, Siying 邓思颖. “汉语句类和语气的句法分析”. In: *汉语学报* 1 (2010), pp. 59–63.
- Dixon, Robert MW. *Basic linguistic theory volume 1: Methodology*. Vol. 1. OUP Oxford, 2009.
- Dixon, Robert MW. *Basic linguistic theory volume 2: Grammatical topics*. Vol. 2. Oxford University Press on Demand, 2010.
- Duan, Xufeng and Cai, Zhenguang G. “Chinese Character Processing”. In: *Oxford Research Encyclopedia of Linguistics*. 2024.
- Grimm, Nadine. *A grammar of Gyeli*. Language science press, 2021.
- He, Leshi 何乐士. *Shǐjì yǔfǎ tèdiǎn yánjiū* 史记语法特点研究 [A study on grammatical features of Historical Records]. Shāngwù Yīnshūguǎn, 2005.
- Huddleston, Rodney and Pullum, Geoffrey K. *The Cambridge Grammar of the English Language*. Cambridge University Press, 2002.
- Jacques, Guillaume. *A grammar of Japhug*. Vol. 1. Language Science Press, 2021.

- Jacques, Guillaume. “On the nature of morphological alternations in Archaic Chinese and their relevance to morphosyntax”. In: *Bulletin of SOAS* 85.3 (2022), pp. 475–494.
- Keidan, Artemij. “Subjecthood in Pāṇini’ s grammatical tradition”. In: *Studi in onore di Giuliano Boccali* (2017), p. 107.
- Kiparsky, Paul. “On the architecture of Pāṇini’ s grammar”. In: *Sanskrit Computational Linguistics: First and Second International Symposia Rocquencourt, France, October 29-31, 2007 Providence, RI, USA, May 15-17, 2008 Revised Selected and Invited Papers*. Springer. 2009, pp. 33–94.
- Kramer, Ruth. “Case study 1: Two genders, three ns”. In: *The Morphosyntax of Gender*. Oxford University Press, Aug. 2015. eprint: <https://academic.oup.com/book/0/chapter/272101874/chapter-pdf/39098307/acprof-9780199679935-chapter-5.pdf>.
- Li, Zuofeng 李佐丰. 古代汉语语法学. 商务印书馆, 2004.
- List, Johann-Mattis et al. “Vowel purity and rhyme evidence in Old Chinese reconstruction”. In: *Lingua Sinica* 3 (2017), pp. 1–17.
- Maling, Joan. “Dative: The heterogeneity of the mapping among morphological case, grammatical functions, and thematic roles”. In: *Lingua* 111.4-7 (2001), pp. 419–464.
- Matthews, Peter Hugoe. *What Graeco-Roman grammar was about*. Oxford University Press, 2019.
- Mazzitelli, Lidia Federica. “From aspect to information structure: Non-aspectual functions of change of state markers in Austronesian and beyond”. In: *Linguistic Typology* 29.1 (2025), pp. 1–33.
- McInnerney, Andrew. “The argument adjunct distinction and the structure of prepositional phrases”. The paper claims that in English the post-verbal constituents have a flat structure. I’m not too sure about that, considering all the work done in Cartography. PhD thesis. University of Michigan, 2022.
- McInnerney, Andrew. “The argument/adjunct distinction does not condition islandhood of PPs in English”. In: *Linguistic Inquiry* (2023), pp. 1–13.
- Mei, Guang 梅广. *Shàngǔ hànǔ yǔfǎ gāngyào* 上古漢語語法綱要 [A sketch of Old Chinese grammar]. Sānmín Shūjú 三民書局, 2015.
- Meisterernst, Barbara. “The syntax of aspecto-temporal adverbs from Late Archaic to Early Medieval Chinese”. In: *Journal of East Asian Linguistics* 25.2 (2016), pp. 143–181.
- Meisterernst, Barbara. “Verb Classes and Aspects: Situation Type in Pre-Tang Chinese”. In: *Zeitschrift der Deutschen Morgenländischen Gesellschaft* 163.1 (2013), pp. 173–202.
- Myers, James. *The grammar of Chinese characters: Productive knowledge of formal patterns in an orthographic system*. Routledge, 2019.
- Pan, Victor Junnan. “Sentence-final particles in Chinese”. In: *Oxford research encyclopedia of linguistics*. Oxford: Oxford University Press. Published online on 28 (2021).
- Paul, Waltraud. *New perspectives on Chinese syntax*. Vol. 271. Walter de Gruyter GmbH & Co KG, 2014.
- Paul, Waltraud. “Why particles are not particular: Sentence-final particles in Chinese as heads of a split CP”. In: *Studia linguistica* 68.1 (2014), pp. 77–115.

- Pinkster, Harm. *The Oxford Latin Syntax: Volume 1: The Simple Clause*. Oxford University Press, Aug. 2015.
- Pulleyblank, Edwin George. *Outline of classical Chinese grammar*. Ubc Press, 1995.
- Punske, Jeffrey. “Three forms of English verb particle constructions”. In: *Lingua* 135 (2013), pp. 155–170.
- Rizzi, Luigi and Cinque, Guglielmo. “Functional categories and syntactic theory”. In: *Annual Review of Linguistics* 2 (2016), pp. 139–163.
- Siddiqi, Daniel. *Syntax within the word: Economy, allomorphy, and argument selection in Distributed Morphology*. Vol. 138. John Benjamins Publishing, 2009.
- Travis, Lisa. “The syntax of reduplication”. In: *PROCEEDINGS-NELS*. Vol. 31. 2. 2001, pp. 455–470.
- Tsai, Wei-Tien Dylan. “Tense anchoring in Chinese”. In: *Lingua* 118.5 (2008), pp. 675–686.
- Vikner, Sten. “Reichenbach revisited: one, two, or three temporal relations?” In: *Acta linguistica hafniensia* 19.2 (1985), pp. 81–98.
- Wiltschko, Martina. *The universal structure of categories*. Vol. 142. Cambridge University Press, 2014.
- Yang, Zhaowei 杨昭蔚, Kong, Lingda 孔令达, and Zhou, Guoguang 周国光. *Gǔhànyǔ cílèi huóyòng cídiǎn 古汉语词类活用词典 (A dictionary of word class changes of Classical Chinese)*. 三环出版社, 1991.
- Zhu, Dexi 朱德熙. “语法讲义”. In: 朱德熙文集, 第一卷. 商务印书馆, 2009.